

A Symmetry Quotient Attack on Odd-Characteristic SNOVA

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Abstract

SNOVA is a multivariate signature scheme that compresses the public key of Unbalanced Oil and Vinegar by representing variables and coefficients with small matrices. After wedge attacks weakened its characteristic-two design, SNOVA adopted odd-prime fields and symmetric public matrices. This paper shows that the symmetry creates a different weakness in the public verifier.

We analyze the common-column relation already used in affine-column forgery attacks. For a symmetric odd-characteristic SNOVA instance whose public matrix algebra has dimension d , symmetry reduces the number of distinct restricted quadratic forms from d^2 to $d(d+1)/2$. Thus sixteen forms become ten when $\ell = 4$, and four become three when $\ell = 2$. An invertible public change of output coordinates then separates these quadratics from the remaining affine constraints and target checks. For every published Version 2.3 parameter shape that passes the stated public rank checks, this gives a quadratic system with the same number of equations and variables. The reduction preserves every verifier equation, rejection rule, and final verification check.

The public reduction is exact, while the recovery estimates are conditional. Under a stated root condition and an unproved regularity hypothesis inherited from symbolic homotopy, the Version 2.3 analysis gives arithmetic exponents 132.188, 183.795, and 233.844 before an unmeasured implementation multiplier. The analysis uses complete systems and, where advantageous, certified smaller systems. Any branch-averaged figure refers to an idealized distribution of public transcripts; public tests select one branch for a fixed key. These are not wall-clock or low-memory bounds. We do not claim a production-size solve or forgery.

1 Introduction

NIST’s Additional Digital Signatures process seeks signature schemes with security and performance profiles that complement the standards selected in its first post-quantum process. SNOVA is one of the nine candidates advanced to the third round in May 2026 [1, 2]. It belongs to the Unbalanced Oil and Vinegar line of multivariate signatures, where verification evaluates a public quadratic map and signing uses a hidden central map with an oil-and-vinegar structure [3, 4].

SNOVA obtains compact public keys by representing variables and coefficients with small matrices and generating many public forms from a smaller structured family. That structure has led to key-recovery and direct forgery attacks. The former expose the hidden UOV structure, while Beullens uses a public relation among signature columns. Later work improves both lines through group actions, extension-field lifting, and multihomogeneous solving [5–10]. These works analyze the original or Round 2 construction over $\mathbb{F}_{16}$.

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The next revision answered Ran’s wedge attack, which uses the alternating polar forms available in characteristic two. Bros et al. adapt the method to SNOVA’s block structure [11, 12]. The SNOVA team responded with odd-prime fields and symmetric public quadratic matrices, while warning that symmetrization could create new attacks [13]. NIST later described this change as a response to the wedge attack and did not regard SNOVA as stable [2].

This change entered the specification in Version 2.1, not Version 2.3. Version 2.3 subsequently added rectangular signatures and revised parameter sets [4]. Our quotient theorem concerns the symmetric odd-characteristic construction and is not specific to those nine parameter sets. The concrete rank, root, and cost analysis covers the nine published $q = 19$ Version 2.3 shapes.

Prior attacks already cover odd characteristic. Jin et al. give a UOV key-recovery attack in arbitrary characteristic [14]. Ran gives an odd-characteristic construction using symmetric forms, but SNOVA’s added structure requires separate analysis [11]. Ding, Li, and Tao give a wedge framework valid in any characteristic, while their concrete SNOVA estimates cover the even-characteristic Round 2 parameters [15]. None combines the symmetric public matrices in the current $q = 19$ construction with the affine-column forgery relation.

This brings us to the question addressed in this paper. What happens when an affine-column attacker imposes the same common-column relation on the new symmetric verifier? We prove that symmetry makes pairs of restricted quadratic forms identical. An invertible public change of output coordinates then separates these quadratics from the remaining affine constraints and target checks. For every published $q = 19$ parameter shape that passes the stated public rank checks, this leaves a quadratic system with the same number of equations and variables.

This statement concerns the restricted verifier, not the unrestricted matrix-valued map. Let $A = \mathbb{F}_q[S]$ have dimension d , and let $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. After all signature columns are expressed through one stacked column u , symmetry gives $q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u)$. The d^2 ordered labels therefore factor through $\text{Sym}^2(A)$, with the exact dimension change

$$d^2 \longrightarrow \binom{d+1}{2}, \quad 16 \longrightarrow 10 \text{ for } d = 4, \quad 4 \longrightarrow 3 \text{ for } d = 2. \quad (1)$$

The full verifier relates the two ordered features by matrix transposition; only the common-column restriction makes them the same scalar polynomial. The identification summarized in Equation (1) is exact and basis-independent, but deliberately local to the attacker-imposed restriction (Theorem 4.1).

1.1 Our results

An exact symmetry quotient and public reduction. After the common-column restriction, every symmetric public base form factors through $\text{Sym}^2(A)$. This loss occurs before the outer public mixing, so later linear combinations cannot restore the missing forms. Public row reduction then separates the surviving quadratics from the affine conditions and target checks. For every published parameter shape that passes the stated public rank checks, this gives an exact square quadratic system. Every recovered root is reconstructed and checked by the original verifier (Theorems 4.1, 4.3 and 4.4).

Conditional recovery and branch-dependent costs. Producing a square system does not prove that it contains an acceptable root or that a solver can find one efficiently. We treat these as separate questions. The root analysis uses either an idealized transcript model or an exact spectrum certificate for one fixed key. The solver analysis assumes an unproved homotopy hypothesis H_{hom} . Under these conditions, the Version 2.3 analysis gives arithmetic exponents 132.188, 183.795, and

233.844 before the unmeasured multiplier κ_{hom} (Theorem 7.1). Exact determinant tests choose between smaller and complete systems. For a fixed key this choice is deterministic. Where both branches contribute, the reported exponent averages them over an idealized distribution of public transcripts. None is a wall-clock or low-memory bound.

Design implications. We derive necessary output-dimension conditions for redesigns that retain the common-column architecture. These conditions are not sufficient for security (Theorems 9.2 and 9.4).

1.2 How to read the result

The paper separates the public reduction, the existence of acceptable roots, and the behavior of the algebraic solver. These statements have different logical status. The following table records the distinction used throughout the paper.

Table 1: Logical status of the attack. An exact statement is a deterministic claim about the supplied public key after its stated public checks pass.

Part	Status	Meaning
Column restriction and symmetry quotient	Exact	Public substitution followed by an algebraic identity.
Output split and affine elimination	Exact after public rank checks	Invertible row reduction preserves every verifier equation.
Frequency of structural and spectrum events	Distributional, or certified for a fixed key	The paper uses an idealized XOF-transcript model; a fixed key instead requires the exact public checks and a spectrum certificate.
Homotopy recovery and cost	Conditional	Completeness and arithmetic cost require H_{hom} and the unmeasured multiplier κ_{hom} .
Candidate acceptance	Exact	Every candidate is reconstructed and checked by the original verifier.
Practical time and memory	Not established	No production solve is given, and the analysis does not bound wall-clock time or peak memory.

1.3 Attack overview

In the hidden UOV coordinates, SNOVA contains no product of two oil blocks. The signer first chooses the vinegar blocks, after which the signing equations are linear in the oil entries. A secret invertible change of variables hides this division, so the public verifier presents a structured quadratic map rather than the linear system used by the signer (Section 2).

The attack seeks a direct public preimage, not the hidden oil space. It starts from the common-column relation used in earlier forgery attacks. Inside every matrix-valued signature block, the attacker writes all columns as public affine functions of one stacked column. This selects a public subset of possible signatures; it is not an identity of the unrestricted verifier (Section 4).

After this restriction, every ordered pair of public matrix multipliers gives a scalar quadratic polynomial. Symmetry makes the polynomials for the two orders equal. One group of sixteen ordered pairs therefore contributes at most ten distinct quadratics when $\ell = 4$, while one group of four contributes at most three when $\ell = 2$. The equality holds before the outer public map combines these forms, so later linear combinations cannot restore the missing quadratic directions (Theorem 4.1).

The attacker next computes an invertible public change of verifier outputs. Some new coordinates contain the surviving quadratic forms. The remaining coordinates are affine conditions on the signature and, when necessary, consistency conditions on the target. Solving the affine equations gives a public affine space. Restricting this space to as many variables as surviving quadratics gives a square system without dropping any verifier equation ([Theorems 4.3 and 4.4](#)).

There are then two recovery choices. The complete system solves every residual quadratic equation on a square slice. A smaller system solves only selected equations on a lower-dimensional slice and checks every omitted equation afterward. The smaller system has a much smaller algebraic state, but a random trial hits the full target less often. An exact public determinant test permits the smaller system; if that test fails, the current algorithm uses the complete system ([Section B.3.1](#)).

The root analysis bounds the chance that a trial contains an accepted root with a full-rank Jacobian. The solver analysis is separate. It starts from a polynomial system with known roots and symbolically deforms that system into the SNOVA residual system. The hypothesis H_{hom} requires the deformation and its specialization to behave regularly enough for the solver to recover the nonsingular roots within the stated arithmetic cost. This hypothesis affects completeness and cost, but not the validity of a returned signature ([Section A](#)).

Finally, every recovered root is converted back into a candidate signature, checked against the public format rule, and passed to the original verifier. A failed rank test, target check, solve, or final verification causes a restart. The final check makes the attack one-sided: it can fail to return a signature, but it never returns an invalid one.

2 The SNOVA signature scheme

This section defines the part of Version 2.3 used in the attack. Fix parameters (v, o, q, ℓ, r) and put $n = v + o$. A vector of private or public variables has the form

$$X = (X_1, \dots, X_n) \in \text{Mat}_{\ell \times r}(\mathbb{F}_q)^n.$$

In the private coordinates, $X_1, \dots, X_v$ are the vinegar blocks and $X_{v+1}, \dots, X_n$ are the oil blocks. The central map has o outputs in $\text{Mat}_{r \times \ell}(\mathbb{F}_q)$, so both its oil input and its output contain $or\ell$ field elements [[3, 4](#)].

The hidden central map. The UOV structure excludes every product of two oil variables. Write $[n] = \{1, \dots, n\}$. The allowed ordered pairs are

$$\Omega = [n]^2 \setminus \{v+1, \dots, n\}^2. \quad (2)$$

Let $S \in \text{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial, and let

$$A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}.$$

Every element of A is a symmetric matrix, and all elements of A commute. Version 2.3 uses

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad N_\alpha = r\ell + r$$

base forms and outer summands. For each $h \in [m_1]$, the secret base form is an n by n block matrix $[F_h] = (F_{h,jk})$, where $F_{h,jk} \in \text{Mat}_\ell(\mathbb{F}_q)$ and $F_{h,jk} = 0$ whenever $(j, k) \notin \Omega$.

We write $L_{i,\alpha}$ and $R_{i,\alpha}$ for the matrices called $A_{i,\alpha}$ and $B_{i,\alpha}$ in the specification. Their dimensions are r by r and r by ℓ , respectively. The other public factors $Q_{i,\alpha,1}, Q_{i,\alpha,2}$ are nonzero elements of A . Put $h(i, \alpha) = 1 + ((i - 1 + \alpha) \bmod m_1)$. The i th component of the hidden central map is

$$\tilde{F}_i(X) = \sum_{\alpha=0}^{N_\alpha-1} L_{i,\alpha} \left(\sum_{(j,k) \in \Omega} X_j^\top Q_{i,\alpha,1} F_{h(i,\alpha),jk} Q_{i,\alpha,2} X_k \right) R_{i,\alpha}, \quad i \in [o]. \quad (3)$$

The inner sum is an r by r matrix, and the full output is an r by ℓ matrix. Since Equation (3) has no product of two oil blocks, fixing the vinegar blocks makes it linear in every entry of the oil blocks.

The secret change of variables and the public map. The secret matrix $T^{12} = (T_{ab})$ has v by o entries in A . It defines the invertible change of variables $X = T(U)$ by

$$\begin{aligned} X_a &= U_a - \sum_{b=1}^o T_{ab} U_{v+b}, \quad 1 \leq a \leq v, \\ X_{v+b} &= U_{v+b}, \quad 1 \leq b \leq o. \end{aligned} \quad (4)$$

Thus T hides the private vinegar and oil coordinates while preserving the matrix shape of each block. For every base form, key generation computes

$$[P_h] = [T]^\top [F_h] [T]. \quad (5)$$

The commutativity of A then gives the public quadratic map

$$\begin{aligned} \tilde{P}_i(U) &= \sum_{\alpha=0}^{N_\alpha-1} L_{i,\alpha} \left(\sum_{j,k=1}^n U_j^\top Q_{i,\alpha,1} P_{h(i,\alpha),jk} Q_{i,\alpha,2} U_k \right) R_{i,\alpha} \\ &= \tilde{F}_i(T(U)). \end{aligned} \quad (6)$$

For odd q , the public base forms are symmetric. In block notation, this means

$$P_{h,kj} = P_{h,jk}^\top. \quad (7)$$

Equivalently, expanding all block entries over $\mathbb{F}_q$ gives a symmetric $n\ell$ by $n\ell$ matrix P_h . This is the symmetry used by our attack.

The block equations also explain the public key compression. Partition the base forms according to the vinegar and oil coordinates:

$$[F_h] = \begin{bmatrix} F_h^{11} & F_h^{12} \\ F_h^{21} & 0 \end{bmatrix}, \quad [T] = \begin{bmatrix} I & -T^{12} \\ 0 & I \end{bmatrix}, \quad [P_h] = \begin{bmatrix} P_h^{11} & P_h^{12} \\ P_h^{21} & P_h^{22} \end{bmatrix}.$$

The public seed determines the outer matrices, the Q matrices, and $P_h^{11}, P_h^{12}, P_h^{21}$. The private seed determines T^{12} . Key generation reconstructs the hidden blocks and the remaining public block by

$$\begin{aligned} F_h^{11} &= P_h^{11}, & F_h^{12} &= P_h^{12} + P_h^{11} T^{12}, \\ F_h^{21} &= P_h^{21} + (T^{12})^\top P_h^{11}, & P_h^{22} &= -(T^{12})^\top P_h^{12} - P_h^{21} T^{12}. \end{aligned} \quad (8)$$

Thus the compressed public key is the public seed together with the blocks P_h^{22} for $h \in [m_1]$. The secret key contains the seeds needed to regenerate T and the hidden forms [4].

Signing and verification. For a message M and a 16-byte salt, the target is

$$Y = H(s_{\text{public}} \parallel \text{SHAKE256}(M, 64) \parallel \text{salt}) \in \text{Mat}_{r \times \ell}(\mathbb{F}_q)^o. \quad (9)$$

The signer derives the vinegar blocks from the private seed, M , the salt, and an attempt counter. Write $V = (X_1, \dots, X_v)$ for the chosen vinegar blocks and let z list the $or\ell$ entries of the oil blocks. By Equation (2), the signing equation is a square linear system

$$G_V z = \text{vec}(Y) - c_V, \quad (10)$$

where $c_V = \text{vec}(\tilde{F}(V, 0))$ and G_V is the coefficient matrix of the terms containing one vinegar block and one oil block. If G_V is singular, the signer chooses a new attempt. Otherwise it solves for the oil blocks, sets $U = T^{-1}(X)$, and retries if U violates the public format rule. The signature is $U \parallel \text{salt}$.

The verifier expands the public key, checks the same format rule, recomputes Y , and accepts exactly when

$$\tilde{P}(U) = Y. \quad (11)$$

Version 2.1 introduced the odd-prime fields and Equation (7). Version 2.3 retained that symmetry and allowed $r \neq \ell$, which gives rectangular signature blocks.

Column form of the public verifier. For each column $c \in \{0, \dots, r-1\}$, stack the c th columns of the public signature blocks as

$$u_c = (U_1[:, c]^T, \dots, U_n[:, c]^T)^T \in \mathbb{F}_q^{n\ell}. \quad (12)$$

Using the scalar expansion of $[P_h]$ and $\Sigma_\xi = I_n \otimes \xi$, each scalar entry inside Equation (6) is a linear combination of terms

$$u_c^T \Sigma_\xi P_h \Sigma_\eta u_d, \quad \xi, \eta \in A. \quad (13)$$

The outer matrices $L_{i,\alpha}$ and $R_{i,\alpha}$ combine these terms by a public linear map. Thus SNOVA's matrix structure affects the scalar verifier through the ordered left and right algebra multipliers in Equation (13). Section 4 imposes the attacker's column relation and determines which of these ordered terms remain distinct.

3 Prior attacks and mathematical background

3.1 SNOVA's attack and revision history

The first line of SNOVA cryptanalysis attacks the hidden UOV structure. Ikematsu and Akiyama analyze key recovery on the scalar expansion of the core map, and Li and Ding sharpen the interpretation as a UOV system with ℓv vinegar variables and ℓo oil variables over the base field [5, 6]. Nakamura, Tani, and Furue instead lift the same structure to a smaller UOV instance over $\mathbb{F}_{q^\ell}$ [10]. These attacks seek the hidden oil space or an equivalent signing key. They concern the original characteristic-two construction and do not use the symmetric public matrices introduced later.

Beullens exposes a different layer. He rewrites SNOVA as a structured MAYO-style whipping of a bilinear map, chooses an affine relation among the signature columns, and searches for rank-deficient combinations in the outer coefficient map [7]. The result is a forgery attack, including weak-key instances much cheaper than the generic case. The Round 2 changes vary the outer coefficient matrices and mitigate the largest rank drops, but they retain the affine-column attack surface.

Our attack starts from this public restriction and acts one layer earlier: symmetry identifies the restricted power-pair polynomials before the outer map sees them.

Cabarcas et al. and Furue et al. improve the algebraic solving of the systems arising in the earlier attacks [8, 9]. The former exploits stability under a commutative group action and adapts the result to reconciliation and direct forgery. The latter writes the extension-field transformation explicitly and applies the resulting multihomogeneous systems to intersection and rectangular-MinRank key recovery. Their estimate for the old Round 2 $(q, v, o, \ell) = (16, 29, 6, 5)$ set is below the claimed Level-V threshold, subject to their stated solving-degree heuristic. Both papers study the characteristic-two construction. Neither derives a loss of independent verifier equations from the later symmetric public matrices.

Ran’s wedge attack begins from the alternating polar forms of characteristic-two UOV and recovers the hidden oil space through exterior algebra [11]. His latest version also gives a different construction for odd characteristic, but states that SNOVA’s added structure prevents the analysis from carrying over directly. Bros et al. specialize the characteristic-two method to SNOVA and give conjectural estimates below the claimed security levels for every Round 2 parameter set [12]. The SNOVA team’s later analysis replaces the conjectural kernel formula with a generating function and finds six Round 2 sets below target [16].

Ding, Li, and Tao place these attacks in a common Macaulay-matrix framework and extend it to multihomogeneous systems in any characteristic [15]. Their formulation applies to SNOVA’s hidden oil-space problem, but their concrete SNOVA analysis is expressly limited to the even-characteristic Round 2 parameters. It does not analyze the symmetric public matrices or the nine $q = 19$ Version 2.3 shapes.

Jin et al. give the main prior attack that genuinely crosses the characteristic boundary [14]. Their p -reduced symmetric algebra recovers Ran’s exterior algebra at $p = 2$ and defines an XL key-recovery attack for arbitrary characteristic. Their concrete odd-characteristic application is to the twelve QR-UOV parameter sets, where the attack does not beat reconciliation on the recommended parameters. The method targets the hidden UOV oil space. It does not analyze the symmetric public-matrix encoding of odd-characteristic SNOVA, and it does not produce the verifier-preserving direct-preimage reduction used here.

The distinction between these attacks is summarized in Table 2. The exact new statement in this paper is the $\text{Sym}^2(A)$ factorization after the public common-column restriction and the resulting invertible change of verifier outputs. The direct-forgery cost estimates built on that reduction remain conditional on their stated root-count and solver hypotheses.

Table 2: Scope of the closest attacks. “Output” is what a successful attack recovers; it is not a claim that every cited estimate is practical.

Work	Targeted regime	Main mechanism	Output and relation to this paper
Ikematsu–Akiyama; Li–Ding; Nakamura– Tani–Furue	Original / early $q = 16$	Scalar or extension-field UOV interpretation	Oil-space or equivalent-key recovery; no odd- q public symmetrization.
Beullens; Cabarcas et al.; Furue et al.	Original and Round 2 $q = 16$	Affine-column rank drop, group action, and multihomogeneous solving	Direct forgery or key recovery; supplies the public restriction that this paper composes with symmetry.
Ran; Bros et al.; Tseng et al.; Ding–Li–Tao	Arbitrary- characteristic UOV frameworks; concrete SNOVA estimates for Round 2 $q = 16$	Exterior or symmetric forms, Macaulay matrices, and the wedge-map kernel	Oil-space recovery; no concrete analysis of the symmetric $q = 19$ SNOVA verifier.
Jin et al.	UOV families in arbitrary characteristic; concrete odd- q QR-UOV instances	p -reduced symmetric-algebra XL	Oil-space recovery; no analysis of symmetric odd- q SNOVA public matrices.
This paper	All nine $q = 19$ Version 2.3 shapes	Common-column restriction, exact $\text{Sym}^2(A)$ quotient, and public output split	Conditional direct preimages checked by the original verifier; no hidden-key recovery claim.

The complementary generic-MQ and partitioning literature includes Hashimoto’s underdefined optimizer, Just Guess, generalized partitioning, and recent F4 engineering [17–20].

3.2 MQ systems, roots, and affine slices

An MQ system over $\mathbb{F}_q$ is a map $g : \mathbb{F}_q^N \rightarrow \mathbb{F}_q^K$ whose coordinates have total degree at most two, together with a target $z \in \mathbb{F}_q^K$. Linear and constant terms are allowed. A root is an x satisfying $g(x) = z$. Underdetermination makes roots more plentiful but does not make the system linear or automatically easy.

An affine space is a translate $a + V$ of a linear subspace. Restricting the search to an affine slice $a + L \subseteq a + V$ turns the problem into a system in $h = \dim L$ coordinates. Three events must be separated:

- (i) the target–slice pair contains a base-field root;
- (ii) it contains an accepted root at which a chosen Jacobian has full rank;
- (iii) the chosen algebraic solver recovers every such root within its stated work bound.

The first two are properties of the map and slice. The third is a solver statement. The later recovery analysis proves them separately.

4 From the public verifier to a square system

This section starts from the column form of the public verifier in Equations (12) and (13) and gives the exact attack as one sequence of public operations. The attacker imposes the common-column relation. Symmetry then identifies restricted quadratic forms. An invertible change of verifier

outputs separates the remaining quadratic equations from the affine ones. Once two public rank checks pass, a final affine restriction leaves equally many quadratic equations and variables. This section makes no claim that the resulting system has a root or that a solver can find one.

4.1 Parameters and public forms

Retain the notation of Section 2 and put

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell+1}{2}. \quad (14)$$

The ceiling follows the pinned Version 2.3 reference implementation [4]. The theorem below proves that K is the maximum number of distinct restricted quadratic forms left by symmetry.

For the public rows, $A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. Write $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$ and use the symmetric scalar expansions $P_1, \dots, P_{m_1} \in \text{Mat}_{N_0}(\mathbb{F}_q)$ from Equation (5). For $\xi, \eta \in A$, define

$$B_i^{\xi, \eta}(x, y) = x^\top \Sigma_\xi P_i \Sigma_\eta y. \quad (15)$$

Setting $x = y$ gives the quadratic power-pair feature.

Table 3: Dimensions of the nine public $q = 19$ Version 2.3 parameter rows.

Level	(v, o, ℓ, r)	m_1	M	K	N_0
I	(28, 5, 4, 4)	5	80	50	132
I	(48, 16, 2, 2)	16	64	48	128
I	(28, 4, 4, 5)	5	80	50	128
III	(40, 7, 4, 4)	7	112	70	188
III	(72, 24, 2, 2)	24	96	72	192
III	(38, 5, 4, 5)	7	100	70	172
V	(50, 9, 4, 4)	9	144	90	236
V	(96, 32, 2, 2)	32	128	96	256
V	(52, 6, 4, 6)	9	144	90	232

4.2 The common-column restriction

Write the stacked signature columns as

$$U = [u_0 \mid \dots \mid u_{r-1}], \quad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses public multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$, and imposes

$$u_j = \Sigma_{R_j} u + v_j, \quad 0 \leq j < r. \quad (16)$$

This relation selects a public subset of possible signatures. It makes no assumption about the honest signer and does not change the public verifier.

Before substitution, the verifier feature vector has $N_{\text{raw}} = m_1 \ell^2 r^2$ labeled coordinates. Let $z_{\text{ord}}(u)$ list the remaining scalar quadratic forms indexed by ordered pairs of powers of S . Let E be the public outer map after the fixed permutation that reconciles the official flattening order with the block-transposed order used here. For a fixed relation R , let R_R expand $z_{\text{ord}}(u)$ into the raw feature vector, and let $F_{R,v}$ collect the coefficients of the terms that are linear in u . The substituted verifier is

$$E(R_R z_{\text{ord}}(u) + F_{R,v} u) + c_{R,v} = t. \quad (17)$$

Put

$$E_R = ER_R, \quad \Lambda_{R,v} = EF_{R,v}.$$

Products of two variable terms give $z_{\text{ord}}(u)$. Products of one variable term and one offset give the linear term, and products of two offsets give the constant.

4.3 How symmetry removes quadratic forms

For one base form in this restricted scalar map, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following theorem identifies the pairs that always give the same polynomial.

Theorem 4.1 (Symmetric-square factorization). *Let $A = \mathbb{F}_q[S]$ have dimension d , and assume $S^\top = S$ and $P_i^\top = P_i$. For each i , the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\text{hom}}, \quad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\text{Sym}_{\mathbb{F}_q}^2(A) = (A \otimes_{\mathbb{F}_q} A) / \text{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1 \binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min \left\{ M, m_1 \binom{d+1}{2} \right\}. \quad (18)$$

If the minimal polynomial of S has degree ℓ , then $d = \ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Every element of A is symmetric. Since the displayed value is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = (u^\top \Sigma_\xi P_i \Sigma_\eta u)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. $\square$

The theorem is an upper bound on the common-column scalar image: extra public dependencies can only reduce that image further. Every concrete route checks the exact public rank before solving. The outer map cannot repair the loss because, after the relation, it sees only the already-equal polynomial values. Before the common-column restriction, swapping (ξ, η) only transposes the matrix $U^\top \Sigma_\xi P_i \Sigma_\eta U$; it does not make the matrix entries equal. The theorem therefore concerns the restricted scalar verifier, not the unrestricted matrix-valued map. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The submitted nonsymmetrized $q = 16$ construction is not covered.

4.4 Mapping the remaining forms to verifier outputs

Let $z_{\text{sym}}(u)$ contain one coordinate for each unordered pair $0 \leq a \leq b < \ell$ in every base form, and let D duplicate its off-diagonal coordinates. By [Theorem 4.1](#), $z_{\text{ord}}(u) = Dz_{\text{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}. \quad (19)$$

The columns of Q_R span every output direction reachable by the quotient quadratics.

4.5 Separating the quadratic and affine equations

Choose W_R whose rows form a basis of the left kernel of Q_R , and put

$$C_{R,v} = W_R \Lambda_{R,v}. \quad (20)$$

Multiplication by W_R removes every quadratic term and exposes the affine constraints. A complementary map G_R keeps the quadratic coordinates. The two maps together are an invertible change of verifier outputs.

Lemma 4.2 (Invertible output split). *Assume $\text{rank } Q_R = K$. Choose a public left inverse $G_R \in \mathbb{F}_q^{K \times M}$ satisfying $G_R Q_R = I_K$. Then*

$$\begin{pmatrix} G_R \\ W_R \end{pmatrix} \in \mathbb{F}_q^{M \times M}$$

is invertible. Writing $b = t - c_{R,v}$, the complete substituted verifier is equivalent to

$$C_{R,v}u = W_R b, \quad g_{R,v}(u) = G_R b, \quad g_{R,v}(u) = z_{\text{sym}}(u) + G_R \Lambda_{R,v} u. \quad (21)$$

The polar map of $g_{R,v}$ is exactly that of z_{sym} . If b is uniform in $\mathbb{F}_q^M$, the two right-hand sides in (21) are independent and uniform.

Proof. If a vector $y \in \mathbb{F}_q^M$ is killed by both G_R and W_R , then $W_R y = 0$ implies $y = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $y = 0$. The stacked square matrix is therefore invertible. Applying its block rows to $Q_R z_{\text{sym}}(u) + \Lambda_{R,v} u = b$ gives the two displayed systems, and invertibility gives the converse. Affine terms have zero polar. Finally, an invertible linear transformation sends a uniform vector to a uniform vector on the product of the two coordinate spaces. $\square$

Proposition 4.3 (Exact affine reduction). *Let $\rho_Q = \text{rank } Q_R$, $\lambda = \text{rank } C_{R,v}$, and $\delta = M - \rho_Q - \lambda$. Then*

$$\text{rank}[Q_R \ \Lambda_{R,v}] = \rho_Q + \lambda.$$

After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ_Q equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as the complete substituted verifier.

If $\rho_Q = K$ and $\lambda = M - K$, every target passes the affine consistency test and the residual system consists of K equations on an affine translate of $\ker C_{R,v}$.

Proof. The row space of W_R is $\ker Q_R^\top$, so $\ker W_R = (\ker Q_R^\top)^\perp = \text{im } Q_R$. Hence W_R realizes the quotient map $\mathbb{F}_q^M \rightarrow \mathbb{F}_q^M / \text{im } Q_R$. The image of the columns of $\Lambda_{R,v}$ in this quotient has dimension $\text{rank}(W_R \Lambda_{R,v}) = \lambda$, proving the first identity.

The projected affine system is $C_{R,v}u = W_R b$. Its codomain has dimension $M - \rho_Q$ and its image has dimension λ , leaving $\delta = (M - \rho_Q) - \lambda$ target consistency equations. Whenever they hold, parameterize the affine solution space by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ_Q rows. Every operation is invertible row reduction or exact affine elimination, so the solution set is unchanged. In the full-rank case, $C_{R,v}$ is surjective and $\delta = 0$. $\square$

4.6 Restricting the remaining equations to a square system

The preceding proposition preserves the full solution set inside the common-column relation. The attack now chooses a smaller affine space on which the number of variables equals the number of remaining equations. This second restriction preserves every verifier equation, but it need not contain a solution. Root existence is handled separately in [Section 6](#).

Corollary 4.4 (Exact square restriction). *Assume*

$$\text{rank } Q_R = K, \quad \text{rank } C_{R,v} = M - K, \quad N_0 \geq M.$$

Then, for every target t , choose one solution $u_\star$ to $C_{R,v}u = W_R(t - c_{R,v})$, a K -dimensional subspace $L \subseteq \ker C_{R,v}$, and an isomorphism $T : \mathbb{F}_q^K \rightarrow L$. On the affine space $u_\star + L$, the complete substituted verifier is equivalent to the square system

$$g_{R,v}(u_\star + Tx) = G_R(t - c_{R,v}), \quad x \in \mathbb{F}_q^K. \quad (22)$$

Every root reconstructs, through (16), a candidate that satisfies the original verifier equations. Conversely, every verifier preimage that satisfies both public restrictions gives a root of (22). The corollary does not assert that a root exists or that it passes the signature-format rejection rule.

Proof. The rank assumption on $C_{R,v}$ makes it surjective, so $u_\star$ exists for every target. Its kernel has dimension

$$N_0 - (M - K) = N_0 - M + K \geq K,$$

so it contains a K -dimensional subspace L . The affine equations in (21) hold throughout $u_\star + L$. The invertible output change in [Theorem 4.2](#) therefore leaves exactly the K quadratic equations in (22). That lemma also gives both directions of the claimed equivalence. $\square$

All nine published parameter shapes satisfy $N_0 \geq M$ by [Table 3](#). Thus, for these shapes, the two displayed rank checks are enough to construct the exact square system. They are not enough to show that this system contains an acceptable root.

Operational meaning. The attacker computes every matrix in this section from the supplied public key. The attacker may test several public relations and offsets before solving, but may not replace an unfavorable key. This paper does not prove how often that search succeeds or measure its cost separately. Those costs belong to κ_{hom} or κ_{JG} for the chosen recovery method. If no tested relation passes the two rank checks, the conditional recovery results do not apply to that key. Once the checks pass, the steps in this section are exact public computations and no verifier equation is dropped.

4.7 A Level-I dimension walkthrough

For $(v, o, \ell, r) = (28, 5, 4, 4)$, we have $n = 33$, $N_0 = 132$, $M = 80$, $m_1 = 5$, and $K = 50$. The common-column relation leaves 80 ordered quadratic labels, but the symmetry quotient leaves only 50 distinct quadratics. In the full-rank affine case, the other 30 output directions become linear equations and the full affine residual domain has dimension $132 - 30 = 102$. The attacker can therefore choose a 50-dimensional affine space and obtain 50 quadratic equations in 50 variables. This is the exact reduction. Whether the chosen space contains an acceptable root, and how much work is needed to find one, are the subjects of the conditional analysis later in the paper.

5 Preparing square systems for recovery

The exact reduction permits any suitable affine slice inside $\ker C_{R,v}$. Recovery needs a slice chosen without reading its fresh coefficients, signatures that pass the format rule, and uniform targets. This section provides those inputs; root existence is addressed in [Section 6](#).

5.1 An unconditional stable kernel

Generic offsets can supply the missing affine output directions, but their linear terms may destroy the block grading used by multihomogeneous solvers. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \dots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \quad (23)$$

Define the offset row space

$$U_w = \text{span}_{\mathbb{F}_q} \{w^\top \Sigma_\xi P_i \Sigma_\eta : 1 \leq i \leq m_1, \xi, \eta \in A\}. \quad (24)$$

For a basis $1, S, \dots, S^{d-1}$ of A , form the complete right- A orbit

$$O_{R,v}^F = \begin{pmatrix} F_{R,v} \\ F_{R,v} \Sigma_S \\ \vdots \\ F_{R,v} \Sigma_{S^{d-1}} \end{pmatrix}, \quad \rho_F = \text{rank } O_{R,v}^F. \quad (25)$$

Theorem 5.1 (Stable-lift kernel). *Assume $S^\top = S$, $P_i^\top = P_i$, and $A \cong \mathbb{F}_{q^d}$. Under (23):*

- (i) U_w is closed under right multiplication by A and $\dim_{\mathbb{F}_q} U_w \leq m_1 d^2$;
- (ii) every row of $F_{R,v}$ and $O_{R,v}^F$ lies in U_w ;
- (iii) the row space of $O_{R,v}^F$ is an A -subspace, so $d \mid \rho_F$;
- (iv)

$$H_{R,v}^F = \ker O_{R,v}^F$$

is A -linear, satisfies $F_{R,v} H_{R,v}^F = 0$, and has

$$\dim_A H_{R,v}^F = n - \rho_F/d \geq n - m_1 d. \quad (26)$$

No generic orbit-rank equality is required. An unexpected dependency only enlarges the stable space.

Right A -stability puts the orbit rows in an A -linear row space, and rank-nullity gives (26). The complete proof is in [Section B.4](#).

Since $C_{R,v} = W_R E F_{R,v}$, the stable kernel lies inside $\ker C_{R,v}$. Translation along a subspace $L \subseteq H_{R,v}^F$ changes the quotient quadratics but leaves every offset-linear feature constant.

5.2 Reserved-line decoupling and the structural preflight

Let $e_1, \dots, e_n$ be the native A -coordinate basis in which the key-dependent symmetric public matrices are expanded. Fix a vinegar index j_0 and use the coordinate-aligned decomposition

$$\begin{aligned} W &= A e_{j_0}, & V' &= \bigoplus_{\substack{j \text{ vinegar} \\ j \neq j_0}} A e_j, & O_{\text{pub}} &= \bigoplus_{j \text{ oil}} A e_j, \\ V_{\text{pub}} &= W \oplus V' \oplus O_{\text{pub}}, & \dim_A W &= 1, & \dim_A V' &= v - 1, & \dim_A O_{\text{pub}} &= o. \end{aligned} \quad (27)$$

The actual SHAKE stream is deterministic once the public seed is fixed. For probability statements only, the *native-coordinate idealized XOF-transcript model* replaces the relevant bytes by independent uniform bytes and reduces the native upper-triangular entries of the key-dependent P_i modulo 19. This is not a claim about a fixed SHAKE-generated key. The model conditions on E , R , and the full vector $\Gamma = (\Gamma_0, \dots, \Gamma_{r-1})$, and invokes independence before any A -linear basis change. A reduced byte has maximum atom

$$\rho_{\text{byte}} = \frac{14}{256},$$

because nine residues have fourteen preimages and ten have thirteen.

Lemma 5.2 (Coordinate-aligned reserved-line decoupling). *Fix E, R, Γ , choose $0 \neq w \in W$, and use the structured offsets (23). Then Q_R depends only on E and R , while $C_{R,v}$, $O_{R,v}^F$, $H_{R,v}^F$, and every deterministically tie-broken slice selected from them depend only on native entries having at least one index in W . In the native-coordinate idealized XOF-transcript model they are independent of the native $V' \times V'$ entries.*

Every offset contains one coordinate from W , so the construction never reads a native $V' \times V'$ entry. The complete proof is in [Section B.4](#).

Lemma 5.3 (Maximum-atom pivot bound). *Let $X_1, \dots, X_N$ be independent finite-field variables, each with maximum atom at most ρ_{byte} . If a matrix B has rank t , then for every c ,*

$$\Pr[BX = c] \leq \rho_{\text{byte}}^t.$$

Conditioning on the nonpivot variables leaves one allowed value for each pivot, so the probabilities multiply. The complete proof is in [Section B.4](#).

Put $\mathcal{Q}_R = \mathbb{F}_q^M / \text{im } Q_R$. When $\text{rank } Q_R = K$, $\dim_{\mathbb{F}_q} \mathcal{Q}_R = M - K$, and $C_{R,v}$ is a coordinate matrix for the induced linear map $u \mapsto W_R \Lambda_{R,v} u$ to $\mathcal{Q}_R$. Let $\mathbf{X}_{W,V'}$ be the vector of native independent public-matrix entries in the $W \times V'$ blocks. For $0 \neq \lambda \in \mathcal{Q}_R^\vee$, define B_λ by

$$\text{vec}((\lambda \circ C_{R,v})|_{V'}) = B_\lambda \mathbf{X}_{W,V'} + b_\lambda.$$

Here B_λ and b_λ are fixed once E, R, Γ, λ and the native layout are fixed; they do not depend on the sampled entries in $\mathbf{X}_{W,V'}$. Scaling λ scales B_λ , so its rank depends only on $[\lambda]$. Put

$$t_{\text{src}} = \min_{0 \neq \lambda \in \mathcal{Q}_R^\vee} \text{rank } B_\lambda.$$

For an A -line $\mathfrak{l} \subseteq W \oplus O_{\text{pub}}$, choose $0 \neq y_{\mathfrak{l}} \in \mathfrak{l}$ and write

$$O_{R,v}^F y_{\mathfrak{l}} = D_{\mathfrak{l}} \mathbf{X}_{\text{nat}} + e_{\mathfrak{l}}$$

in the complete vector $\mathbf{X}_{\text{nat}}$ of native independent public-matrix entries. The coefficient data $D_{\mathfrak{l}}, e_{\mathfrak{l}}$ are deterministic before those entries are sampled, and the rank of $D_{\mathfrak{l}}$ is independent of the chosen generator. Define

$$t_W = \text{rank } D_W, \quad t_{\text{oth}} = \min_{\substack{\mathfrak{l} \subseteq W \oplus O_{\text{pub}} \\ \dim_A \mathfrak{l} = 1, \mathfrak{l} \neq W}} \text{rank } D_{\mathfrak{l}}.$$

These are exact symbolic coefficient-map ranks determined by the fixed relation, the complete offset pattern, and the native coordinate layout; they do not depend on the sampled values of the key coefficients. Define

$$\delta_{\text{src}} = \frac{q^{M-K} - 1}{q - 1} \rho_{\text{byte}}^{d(v-1)}, \tag{28}$$

$$\delta_{\text{proj}} = \rho_{\text{byte}}^{m_1 \binom{d+1}{2}} + \left(\frac{|A|^{1+o} - 1}{|A| - 1} - 1 \right) \rho_{\text{byte}}^{m_1 d^2}, \quad |A| = q^d. \tag{29}$$

Theorem 5.4 (Conditional coefficient-preflight density). *Fix the deterministic outer map E , a relation R , and the complete vector Γ . Assume the exact deterministic checks*

$$\text{rank } Q_R = K, \quad t_{\text{src}} \geq d(v-1), \quad t_W \geq m_1 \binom{d+1}{2}, \quad t_{\text{oth}} \geq m_1 d^2$$

pass. In the native-coordinate idealized XOF-transcript model,

$$\Pr[\text{rank } C_{R,v} < M - K] \leq \delta_{\text{src}},$$

and

$$\Pr[H_{R,v}^F \cap (W \oplus O_{\text{pub}}) \neq 0] \leq \delta_{\text{proj}}.$$

Consequently full affine rank and injectivity of $H_{R,v}^F \rightarrow V'$ hold jointly with probability at least $1 - \delta_{\text{src}} - \delta_{\text{proj}}$. For the six official $\ell = 4$ dimension tuples, direct substitution gives $\delta_{\text{src}} + \delta_{\text{proj}} < 2^{-209}$. The coefficient-map ranks are fixed-layout symbolic checks; the realized quotient, affine, and intersection ranks are exact public checks on a fixed key.

The two bounds apply [Theorem 5.3](#) to projective output directions and to A -lines in $W \oplus O_{\text{pub}}$, then take union bounds. The complete proof is in [Section B.4](#).

The concrete route records all r entries of Γ and runs the displayed checks. This release does not assert that every fixed key passes them; the six $\ell = 4$ ledgers are conditional on a successful structural preflight.

5.3 The guaranteed fresh ordinary square

Let

$$D_0 = V' \cap \ker C_{R,v}.$$

The intersection inequality gives

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \quad (30)$$

For all six $\ell = 4$ rows this lower bound is at least K :

Parameters	$\dim D_0$ lower bound	K
$(28, 5, 4, 4), (28, 4, 4, 5)$	78	50
$(40, 7, 4, 4)$	114	70
$(38, 5, 4, 5)$	118	70
$(50, 9, 4, 4)$	142	90
$(52, 6, 4, 6)$	150	90

Consequently the attacker can choose, from cross data alone, an ordinary K -dimensional slice inside D_0 . Its internal $V' \times V'$ coefficients remain fresh for the spectrum theorem. This simple dimension fact is the reason the complete-square route applies to every $\ell = 4$ row, including $(38, 5, 4, 5)$; it does not require the smaller structured stable kernel to have A -dimension $K/4$.

5.4 Passing the block-symmetry rejection rule

Version 2.3 rejects a square $\ell = 4$ signature if any block is symmetric and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [4]. Rectangular $\ell = 4$ blocks are not subject to the square-block rule.

For a square $\ell = 4$ relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathcal{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \quad \mathcal{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c.$$

For any offsets, one signature block is symmetric exactly when $\mathcal{S}_R(x)$ equals a fixed offset-dependent vector. The public relation $R_j = S^j$ for the official matrix has rank $\mathcal{S}_R = 4$. Thus each block has at most one rejected common-column value, and the accepted density in the full common-column space is at least

$$\alpha_{4,n} = (1 - 19^{-4})^n. \quad (31)$$

The accepted-root theorem later uses only this global density; it assumes no independence from the MQ system.

For $\ell = 2$, the zero-offset relation used in the main theorem gives a block

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

which is symmetric exactly when $y = 0$. The exact accepted density is derived in (57). A deterministic fixed-skew alternative is recorded in Section B.4; it is not used by the main theorem.

5.5 Exactly uniform official targets

The probability theorems require a uniform target over $\mathbb{F}_{19}^M$, but the official binary-to-field conversion has a small modulo bias. Model the target hash/XOF chunks as independent uniform bit strings, separately from the key expansion above. The pinned conversion uses each full eight-byte chunk for fifteen base-19 digits and a final partial chunk for the remainder. If a uniform b -bit chunk X supplies a digits, retain the message–salt input only when

$$X < \left\lfloor \frac{2^b}{19^a} \right\rfloor 19^a. \quad (32)$$

Conditioned on this event, $X \bmod 19^a$ is uniform. Applying the test to each chunk makes the retained official target exactly uniform, and an invertible output change preserves uniformity. Exact evaluation over the nine public Version 2.3 lengths gives a minimum acceptance probability $0.1524622985\dots$, and hence at least 0.152462 . When target consistency requires more than the 128-bit salt space, the existential chosen-message attacker varies a message prefix as well as the salt. Target generation occurs before nonlinear solving and adds to, rather than multiplies, the solve cost.

6 Root existence on a square slice

The preceding sections produce a square MQ system but do not show that it has a root with the required format and a nonsingular Jacobian. This section derives that probability from an exact collision identity and bounds the required derivative ranks for the slices used later.

6.1 Polar forms, rank spectra, and collisions

For a scalar quadratic polynomial f , define the polar form

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic this is bilinear. For a vector-valued map g , a nonzero $b \in \mathbb{F}_q^K$ selects the scalar quadratic $b \cdot g$. Its polar rank measures how many independent directions the quadratic part couples.

For a fixed slice direction y , the vector-valued derivative

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0) \quad (33)$$

is linear in x . The value $q^{-\text{rank } T_y}$ is exactly the fraction of its codomain-dual directions that annihilate T_y , after the appropriate normalization. Summing these quantities over nonzero y controls the second factorial moment of the number of roots in a random target-coset pair. The same spectrum controls the density of points where the restricted Jacobian loses rank.

6.2 The exact first and second moments

Theorem 6.1 (Root-collision identity). *Let U be an $\mathbb{F}_q$ -space, let $C : U \rightarrow \mathbb{F}_q^c$ be surjective, let $L \subseteq \ker C$ have dimension h , and let $g : U \rightarrow \mathbb{F}_q^K$ have degree at most two. Sample independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, then choose a uniform affine coset $\mathcal{A}$ of L inside $C^{-1}(d)$. Put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \quad \mu = q^{h-K}.$$

For $0 \neq y \in L$, define

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0)$$

and let $\kappa(y)$ be one if $-(g(y) - g(0)) \in \text{im } T_y$ and zero otherwise. Then

$$\mathbb{E}Z = \mu, \quad (34)$$

$$\mathbb{E}[Z(Z - 1)] = \mu \sum_{0 \neq y \in L} \kappa(y) q^{-\text{rank } T_y} \leq \mu \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (35)$$

The domain of T_y is the full space U because the experiment averages over the affine right-hand side as well as the coset.

Proof. Sampling d and then a uniform L -coset in $C^{-1}(d)$ is exactly uniform over all cosets of L in U . Every point of U therefore lies in the sampled coset with probability $q^{h-\dim U}$, and its image matches the independent target with probability q^{-K} . Summing over all points gives (34).

For an ordered pair of distinct roots in one coset, write the second point as $x + y$ with $0 \neq y \in L$. The collision condition is

$$g(x + y) = g(x) \iff T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y) q^{\dim U - \text{rank } T_y}$ solutions in x . Divide the total number of ordered collisions by the number $q^{\dim U - h} q^K$ of target-coset pairs to obtain (35). $\square$

6.3 Accepted nonsingular roots

Retain U, C, g, L and the trial from [Theorem 6.1](#), and write $g = q_2 + q_1 + q_0$ by homogeneous degree. For the full-domain derivative T_y from (33), define

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} q^{-\text{rank } T_y}. \quad (36)$$

The superscript records that the trial averages over the affine right-hand side and the L -coset. The derivative therefore has domain U , not $\ker C$. Restricting it to $\ker C$ gives a different, generally larger quantity used for fixed-fiber uniqueness, not for this experiment.

Let $X \subseteq U$ be the assignments accepted by the original verifier and put $\alpha = |X|/|U|$. A trial samples independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, and then a uniform affine coset of L inside $C^{-1}(d)$.

Theorem 6.2 (Accepted nonsingular root, full-domain form). *Assume $\bar{\eta}_L^U \leq \eta$ and $\alpha > \eta/(q-1)$. A trial contains an accepted root x satisfying*

$$\text{rank } D_L g(x) = h, \quad D_L g(x)[y] = T_y(x) + q_1(y),$$

with probability at least

$$p_{\text{ns}} = q^{h-K} \frac{a^2}{a + \eta}, \quad a = \alpha - \frac{\eta}{q-1}. \quad (37)$$

No independence among acceptance, singularity, the target, and the quadratic map is assumed.

Proof. Let Y count accepted roots in the sampled target-coset pair at which $D_L g$ has rank h , and let Z count all roots. For a projective direction $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is T_y . It therefore holds on at most a $q^{-\text{rank } T_y}$ fraction of U . Union-bound over projective directions to obtain

$$\frac{|\{x : \text{rank } D_L g(x) < h\}|}{|U|} \leq \frac{1}{q-1} \sum_{0 \neq y \in L} q^{-\text{rank } T_y} \leq \frac{\eta}{q-1}.$$

Every point of U is selected by the random fiber-coset with probability $q^{h-\dim U}$ and matches the independent target with probability q^{-K} . Put $\mu = q^{h-K}$ and $a = \alpha - \eta/(q-1)$. Hence $\mathbb{E}Y \geq \mu a$. The ordered-pair collision identity gives $\mathbb{E}[Z(Z-1)] \leq \mu\eta$, and $Y(Y-1) \leq Z(Z-1)$. Therefore

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \leq \mathbb{E}Y + \mu\eta.$$

The function $x \mapsto x^2/(x + \mu\eta)$ is increasing for $x \geq 0$. Cauchy-Schwarz therefore gives

$$\Pr[Y > 0] \geq \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y + \mu\eta} \geq \mu \frac{a^2}{a + \eta},$$

which proves (37). The denominator is strictly smaller than $1 + \eta$ whenever $a < 1$. $\square$

The following lemma bounds this spectrum in the idealized transcript model. Let $\rho_{\text{byte}} = 14/256$ be the maximum atom of one public-expansion byte reduced modulo 19.

Lemma 6.3 (Full-domain fresh-spectrum theorem). *Assume q is odd and $A = \mathbb{F}_{q^d}$. Work in the native-coordinate idealized XOF-transcript model of [Section 5](#): the scalar entries in the fresh symmetric-matrix region are independent reductions of independent uniform bytes, each having maximum atom $\rho_{\text{byte}} = 14/256$, while all disjoint coefficient data are conditioned upon. Let L be*

fixed from coefficient data disjoint from that fresh region. Suppose every $0 \neq y \in L$ has nonzero projection $\bar{y}$ to a fresh A -space D , and every nonzero residual-output combination contains a nonzero coefficient $c \in \text{Sym}_{\mathbb{F}_q}^2(A)$ in at least one fresh public base form. If

$$\dim_{\mathbb{F}_q} D = d + t,$$

then for all $0 \neq y \in L$ and $0 \neq b \in \mathbb{F}_q^K$,

$$\Pr[b \in \ker T_y^\top] \leq \rho_{\text{byte}}^t. \quad (38)$$

Consequently

$$\mathbb{E}[\bar{\eta}_L^U] \leq (q^h - 1)q^{-K} + (q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t, \quad (39)$$

$$\Pr[\bar{\eta}_L^U > (q^h - 1)q^{-K} + \varepsilon] \leq \frac{(q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t}{\varepsilon}. \quad (40)$$

For the reserved-line construction one may take $D = V' \cong A^{v-1}$ and $t = d(v-2)$. For the zero-offset construction on the full public-vinegar summand one may take $D \cong A^v$ and $t = d(v-1)$.

For fixed $y \neq 0$ and $b \neq 0$, the fresh coefficient map has rank at least t . The maximum-atom bound gives (38); expectation and Markov's inequality give (39) and (40). The complete proof is in Section B.4.

For the rest of the paper, use the threshold

$$\boxed{\eta_{128}(h, K) = (q^h - 1)q^{-K} + 2^{-128}.} \quad (41)$$

This allows 2^{-128} above the exact full-rank baseline; it does not assume perfect rank.

Corollary 6.4 (Numerical spectrum tails). *For the cost-driving slices, (40) with $\varepsilon = 2^{-128}$ gives the following base-two failure exponents. The entry in each cell is the weaker official shape when two shapes share a level.*

Route	Level I	Level III	Level V
$\ell = 4$ fast channel core	−138.12	−237.86	−371.14
$\ell = 4$ direct complete square	−95.64	−178.39	−294.67
$\ell = 2$ one-eigenblock core	−130.17	−263.46	−396.74
$\ell = 2$ complete two-block core	−62.21	−161.50	−260.80

Thus, in that idealized transcript model, even the weakest headline event fails with probability below 2^{-62} . A fixed key can instead publish its exact projective rank histogram, which certifies (41) without the idealized XOF-transcript model.

7 Conditional recovery for all published parameter sets

The exact reduction produces square systems for every published parameter shape. Recovery also needs roots with the right format and a solver that finds them. We state those conditions once, then give the resulting costs. The complete-system arithmetic and exact fallback rules are given in Section B.

Assumptions. For each invoked route, the fixed public key and chosen common-column relation must pass the exact quotient, affine-rank, and applicable structural checks. For the $\ell = 2$ route, the attacker also retains only targets that pass the exact public consistency test. Each invoked slice must satisfy $\bar{\eta}_L^U \leq \eta_{128}(h, K)$, either under the idealized transcript model or through an exact spectrum certificate for the fixed key (Section 6). An exact public determinant test selects a smaller system when available; if it fails, the algorithm uses the complete square, so passing this test is not an extra assumption on the key. The table averages the work of the smaller system and its fallback. This average also uses the idealized XOF-transcript bound on how often the structural and determinant tests fail. For a fixed key, the exact test chooses one branch, but the averaged exponent need not describe that key. Finally, H_{hom} must hold for every square system that is solved. The displayed work is the arithmetic ledger multiplied by the unmeasured factor κ_{hom} .

Theorem 7.1 (Conditional recovery for all published parameter shapes). *Consider any published $q = 19$ Version 2.3 parameter shape under the assumptions above. A randomized algorithm solves the certified smaller system when its exact determinant test passes and otherwise solves the complete square. It checks every candidate against all residual equations, confirms that its coordinates lie in the base field, reconstructs the signature, applies the signature rejection rule, and runs the original verifier. It therefore never returns an invalid signature.*

The table gives the reported arithmetic exponents before the unmeasured multiplier κ_{hom} .

Level	parameters	certified route	exponent	headroom
I	(28, 5, 4, 4)	smaller channel system, else full square	128.246	14.754
I	(48, 16, 2, 2)	one eigenblock, else two-block square	132.188	10.812
I	(28, 4, 4, 5)	smaller channel system, else full square	128.246	14.754
III	(40, 7, 4, 4)	smaller channel system, else full square	171.617	35.383
III	(72, 24, 2, 2)	complete two-block square	183.795	23.205
III	(38, 5, 4, 5)	smaller channel system, else full square	171.616	35.384
V	(50, 9, 4, 4)	smaller channel system, else full square	214.558	57.442
V	(96, 32, 2, 2)	complete two-block square	233.844	38.156
V	(52, 6, 4, 6)	smaller channel system, else full square	214.557	57.443

Taking the worst row at each level gives

$$\boxed{132.18807, \quad 183.79463, \quad 233.84399}. \quad (42)$$

These are multiplication-schedule estimates, not wall-clock or low-memory bounds. The theorem does not assert that its conditions hold for every official key, and the artifact does not run a production-size nonlinear solver.

Proof. The accepted-root theorem of Theorem 6.2 gives the root probability under the stated spectrum condition. The complete $\ell = 4$ and $\ell = 2$ systems are proved in Sections B.2.1 and B.2.2. The exact determinant tests, smaller systems, and complete-square fallback are proved in Sections B.3 and B.3.1. The final checks rule out invalid outputs. Substituting the nine published parameter shapes into those bounds gives the table; the exact arithmetic rows appear in Section F. $\square$

8 Implementation, evidence, and reproducibility

The companion artifact provides formula checkers, one official-key reduction test, and one toy end-to-end forgery test. Table 4 states what each establishes. None executes a production-size nonlinear solver.

Table 4: Evidence shipped with the release. “Included” means that the named script and its pinned input are present, not that the headline recovery cost has been measured.

Evidence	Availability	What it establishes
Formula and ledger checks	Included	Numerical consequences of the dimensions, probability models, and cost conventions supplied to the scripts; not scheme correspondence or solver execution.
Scalar circuit netlist	Included	The emitted 150-gate $\mathbb{F}_{19}$ multiplication netlist is correct on all 361 valid inputs. The extension-field figures are recurrence counts, not emitted netlists.
Official KAT verification and reduction	Included for one Level-I (28, 5, 4, 4) key	A Python transcription parses the compressed public key, accepts the supplied signature, rejects an altered message, and obtains quotient/affine ranks 50/30. The resulting 50-equation system in 102 variables is left unsolved.
Reduced zero-offset forgery	Included at unofficial (2, 1, 2, 2)	The complete restricted map is interpolated; its image equals the explicit rank-three quotient image, leaving one target-consistency coordinate before public slice search. The main case, eight committed fresh-key regressions, and a separate 24-forgery/100-rank stress check pass in the shared KAT-anchored Python transcription. This is toy composition evidence, not independent C-verifier or production-cost evidence.
All-shape official-key preflight census	Not included	No empirical pass probability for the fixed-layout or realized-rank hypotheses.
Production-size solver or forgery	Not included or claimed	No empirical evidence for H_{hom} , κ_{hom} , κ_{JG} , memory, or feasibility at a published shape.

Both correspondence tests use a KAT-anchored Python transcription. The official-key test compares the substituted map with a separate literal evaluator but stops before the nonlinear solve. The toy forger reconstructs and serializes a signature and checks the rejection rule, but its forger and verifier share scheme helpers and do not call the official C verifier. It is therefore composition evidence rather than an independent implementation.

The release neither tests H_{hom} nor measures κ_{hom} , memory, or full gate cost. The AXN ledger charges the stated multiplication schedule; control, data movement, indexing, factorization, communication, and peak state remain outside it. [Section A](#) records the solver coordinates and homotopy adaptation. [Section C](#) leaves κ_{JG} unmeasured, while [Section D](#) does not prove its production solving degree.

A production fixed-key record would need the exact preflights, a certified slice spectrum, the complete solver and filtering transcript, the reconstructed signature, and the original verifier transcript. Only an instrumented run that records the omitted work could measure total work and memory.

9 Necessary dimension floors within the common-column model

We give two necessary output floors for redesigns that retain the common-column architecture. The first removes a square slice from every target fiber. The stronger second floor removes the corresponding zero-offset chosen-message slice. Neither floor is sufficient for security.

Proposition 9.1 (Dimensionally square common-column slice bound). *Let $A = \mathbb{F}_{q^d}$ and retain the public vinegar summand V' of base-field dimension $d(v - 1)$. Suppose an exact affine reduction has*

M' outputs, K' quotient quadratics, and full affine rank $M' - K'$. Then

$$\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,v}) \geq d(v-1) - (M' - K').$$

If $M' \leq d(v-1)$, dimensions alone leave an ordinary K' -dimensional slice carrying all K' residual equations. Equality of equation and variable counts does not establish a root, zero-dimensionality, nonsingularity, or an efficient solve. Eliminating this dimensional guarantee requires $M' > d(v-1)$. Under the coupling $M' = dx'$, this forces $x' \geq v$ and $M' \geq dv$.

Proof. Use

$$\dim(V' \cap \ker C_{R,v}) \geq \dim V' - \text{rank } C_{R,v} = d(v-1) - (M' - K').$$

The right side is at least K' exactly when $M' \leq d(v-1)$. Negate and round to the next multiple of d . $\square$

Let $K_{\text{cc,ord}} = \min\{M, m_1 d^2\}$ be the pre-symmetrization ordered-label cap within the same common-column model. Retaining symmetric forms requires $m'_1 \geq \lceil K_{\text{cc,ord}} / \binom{d+1}{2} \rceil$ and $M' \geq K_{\text{cc,ord}}$. Under $M' = dx'$ and $m'_1 = \lceil x'/d \rceil$, all three necessities combine as follows.

Theorem 9.2 (Combined necessary common-column output floor). *Any redesign that retains symmetric public forms, the public vinegar count, the public-vinegar affine geometry above, full affine rank $M' - K'$, and the Version 2.3 output/base-form coupling, while both restoring $K_{\text{cc,ord}}$ and eliminating the dimensionally guaranteed fresh slice, must satisfy*

$$x' \geq \max \left\{ v, \left\lceil \frac{K_{\text{cc,ord}}}{d} \right\rceil, d \left(\left\lceil \frac{K_{\text{cc,ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}, \quad M' = dx'. \quad (43)$$

Across the nine official rows, this requires 45.00%–60.71% more outputs ([Theorem 9.4](#)).

Proof. The second and third terms in (43) are the two ordered-cap requirements under $m'_1 = \lceil x'/d \rceil$. The term v is [Theorem 9.1](#). Their maximum is therefore necessary. Substitution gives the affine-fiber column in the comparison below. $\square$

Zero-offset targets. The preceding floor assumes a full-rank affine lift. A zero-offset attacker can instead filter chosen-message targets through the quadratic image and use the full common-column domain. Removing this square slice requires a stronger floor.

Proposition 9.3 (Zero-offset dimensionally square restriction). *Consider any symmetric redesign for which the zero-offset common-column relation with $R_0 = 1$ is defined. Let Q_R be its quadratic output matrix and put $\rho_Q = \text{rank } Q_R$. Public row reduction gives an invertible output change under which the zero-offset verifier is equivalent to*

$$g_{\rho_Q}(u) = z \in \mathbb{F}_q^{\rho_Q}, \quad t_{\text{con}} = 0 \in \mathbb{F}_q^{M' - \rho_Q}, \quad (44)$$

where g_{ρ_Q} consists of ρ_Q independent quadratic output combinations. For an exactly uniform target, the consistency event in (44) has probability $q^{-(M' - \rho_Q)}$ and, conditioned on it, z is uniform in $\mathbb{F}_q^{\rho_Q}$.

With zero offsets no reserved line is needed, so use the full public-vinegar space $V_{\text{vin}} \cong A^v$, of base-field dimension dv . If

$$\rho_Q \leq dv, \quad (45)$$

then dimensions alone leave an ordinary ρ_Q -dimensional slice $L \subseteq V_{\text{vin}}$ carrying all ρ_Q residual equations: it has ρ_Q equations in ρ_Q variables. This dimension equality alone proves neither root existence nor efficient recovery. Consequently, ruling out this chosen-message dimensional configuration requires

$$\rho_Q > dv. \quad (46)$$

Proof. With zero offsets, the substituted verifier is $Q_R z_{\text{sym}}(u) = t$ and contains no affine or constant term. Choose a row basis of Q_R and extend it to an invertible basis change on the M' output coordinates. The first ρ_Q transformed rows are the independent residual quadratics g_{ρ_Q} ; the remaining rows vanish identically on the quadratic image and therefore impose the consistency equations in (44). An invertible linear transformation sends a uniform target to independent uniform residual and consistency coordinates, proving the distribution statement. Under (45), choose any ρ_Q -dimensional linear subspace of V_{vin} and restrict all ρ_Q residual equations to an affine coset of it. This gives the asserted dimensionally square restriction. $\square$

Symmetry bounds the actual quotient rank by

$$\rho_Q \leq \min \left\{ M', m'_1 \binom{d+1}{2} \right\}. \quad (47)$$

Hence making (46) even possible forces both terms on the right of (47) above dv .

Theorem 9.4 (Output floor eliminating the zero-offset dimension guarantee). *Retain symmetry, the public vinegar count, and the Version 2.3 coupling*

$$M' = dx', \quad m'_1 = \left\lceil \frac{x'}{d} \right\rceil.$$

Put

$$C_d = \binom{d+1}{2}, \quad m_{\text{sq}} = \left\lfloor \frac{dv}{C_d} \right\rfloor + 1.$$

A necessary condition for eliminating the zero-offset chosen-message dimension guarantee is

$$x' \geq \max \{v + 1, d(m_{\text{sq}} - 1) + 1\}, \quad M' = dx'. \quad (48)$$

If the redesign also restores the common-column ordered-label cap $K_{\text{cc,ord}}$, the right side of (48) must additionally be maximized with

$$\left\lceil \frac{K_{\text{cc,ord}}}{d} \right\rceil, \quad d \left(\left\lceil \frac{K_{\text{cc,ord}}}{C_d} \right\rceil - 1 \right) + 1.$$

For the nine official rows, the zero-offset condition dominates the ordered-cap terms. The two repair floors compare as follows.

Level	parameters	current M	necessary outputs M'	
			affine fiber	zero offset
<i>I</i>	(28, 5, 4, 4)	80	116 (45.00%)	180 (125.00%)
<i>I</i>	(48, 16, 2, 2)	64	96 (50.00%)	130 (103.125%)
<i>I</i>	(28, 4, 4, 5)	80	116 (45.00%)	180 (125.00%)
<i>III</i>	(40, 7, 4, 4)	112	180 (60.71%)	260 (132.14%)
<i>III</i>	(72, 24, 2, 2)	96	144 (50.00%)	194 (102.08%)
<i>III</i>	(38, 5, 4, 5)	100	152 (52.00%)	244 (144.00%)
<i>V</i>	(50, 9, 4, 4)	144	228 (58.33%)	324 (125.00%)
<i>V</i>	(96, 32, 2, 2)	128	192 (50.00%)	258 (101.5625%)
<i>V</i>	(52, 6, 4, 6)	144	228 (58.33%)	324 (125.00%)

The zero-offset floor requires 101.5625%–144.00% more outputs.

Proof. Condition (46) and the two caps in (47) require $dx' > dv$ and $m'_1 C_d > dv$. The first inequality is equivalent to $x' \geq v + 1$. The second requires $m'_1 \geq m_{\text{sq}}$, and the least integer x' with $\lceil x'/d \rceil \geq m_{\text{sq}}$ is $d(m_{\text{sq}} - 1) + 1$. This proves (48). The two extra ordered-cap terms are the same feature and output necessities used in Theorem 9.2. Direct substitution gives the table. $\square$

The stronger result is a dimension-only theorem. A redesign below the table can retain the square system and rely on the consistency factor $q^{M' - \rho Q}$, its solving cost, or a changed target interface. The floor applies only if the redesign instead removes the square slice while preserving the common-column architecture and parameter coupling.

10 Conclusion

The common-column restriction exposes the central weakness of the $q = 19$ Version 2.3 construction. Symmetry collapses its restricted quadratic coordinates to a symmetric-square quotient. Conditional on the stated public ranks, an invertible output change separates this quotient from the affine and target-consistency constraints. The result is a square quadratic system that preserves every verifier equation, the rejection rule, and final verification. This statement concerns the restricted verifier, not the unrestricted matrix-valued map.

All nine published shapes meet the dimension requirements of the exact square restriction (Theorem 4.4). Applying the recovery algorithms to a key also requires the stated public preflights (Theorem 7.1). The structural reduction is exact; the homotopy and Just Guess estimates require their stated hypotheses and unmeasured multipliers. The artifact contains no production-size solve or forgery (Table 4).

The repair bounds apply only to designs that retain the common-column architecture and current parameter coupling. They show that restoring the ordered-label count does not by itself remove the stronger zero-offset slice (Theorems 9.2 and 9.4).

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A Conditional solver coordinates and finite-field homotopy

The exact reduction in Section 4 needs only the dimension of the symmetric square. The conditional recovery algorithms use the more structured coordinates and finite-field homotopy adaptation recorded here.

A.1 Frobenius coordinates for the symmetric square

For $A = \mathbb{F}_{q^d}$ and $j > 0$, define

$$\phi_j([x \otimes y]_{\text{sym}}) = xy^{q^j} + x^{q^j}y, \quad \phi_0([x \otimes y]_{\text{sym}}) = xy.$$

If d is even and $j = d/2$, the value lies in $\mathbb{F}_{q^{d/2}}$.

Theorem A.1 (Frobenius-distance normal form). *There is an $\mathbb{F}_q$ -linear isomorphism*

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^d}) \cong \begin{cases} A^{(d+1)/2}, & d \text{ odd}, \\ A^{d/2} \oplus \mathbb{F}_{q^{d/2}}, & d \text{ even}, \end{cases}$$

given by the maps ϕ_j for the distinct Frobenius distances. In particular,

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^4}) \cong \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^2}, \quad (49)$$

$$\text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_{q^2}) \cong \mathbb{F}_{q^2} \oplus \mathbb{F}_q. \quad (50)$$

Proof. Extend scalars to an algebraic closure and identify $A \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ with a product of d copies indexed by $\mathbb{Z}/d\mathbb{Z}$. Frobenius cyclically shifts the primitive idempotents. The map ϕ_0 records the d diagonal unordered pairs. For $0 < j < d/2$, the conjugates of ϕ_j record the orbit of unordered pairs $\{i, i+j\}$, an orbit of size d . If d is even and $j = d/2$, the pair is repeated after a half-turn, leaving an orbit of size $d/2$. These orbits partition all unordered pairs. The displayed map is therefore a coordinate permutation after scalar extension and hence an isomorphism over $\mathbb{F}_q$. $\square$

For $\ell = 4$, the three coordinate groups have highest homogeneous forms of ordinary degrees 2, $q+1 = 20$, and $q^2+1 = 362$ over $A = \mathbb{F}_{19^4}$. For $\ell = 2$, the quotient becomes two groups with bidegrees $(2, 0)$, $(1, 1)$, and $(0, 2)$ after scalar extension.

A.2 Finite-cardinality symbolic homotopy

The selected-degree-free branch adapts a multihomogeneous homotopy algorithm. The scope of the adaptation is important.

Partition $N = \sum_{j=1}^b n_j$ variables into b blocks $X_1, \dots, X_b$. Let $f_1, \dots, f_N$ be nonconstant polynomials represented by a straight-line program of length L , and suppose f_i has multidegree at most $\mathbf{d}_i = (d_{i,1}, \dots, d_{i,b})$. Define

$$B = C_{\mathbf{n}}(\mathbf{d}) = [\vartheta_1^{n_1} \cdots \vartheta_b^{n_b}] \prod_{i=1}^N \left(\sum_{j=1}^b d_{i,j} \vartheta_j \right), \quad (51)$$

$$B^+ = C'_{\mathbf{n}}(\mathbf{d}') = \text{coeffsum} \left(\prod_{i=1}^N \left(\vartheta_0 + \sum_{j=1}^b d_{i,j} \vartheta_j \right) \bmod \langle \vartheta_0^2, \vartheta_1^{n_1+1}, \dots, \vartheta_b^{n_b+1} \rangle \right). \quad (52)$$

Inherited homotopy hypothesis H_{hom} . For the input system and the chosen random parameters, assume that the incidence, deformation-curve regularity, specialization, and cleaning hypotheses used by Proposition 5, Algorithm 2, and Remark 15 of Safey El Din and Schost remain valid after the two finite-field substitutions made below [21]. In particular, the relevant homotopy components must have the expected dimensions and the specializations and inversions used during Newton lifting must avoid their exceptional loci. This paper proves the start-system and separator substitutions below; it does not derive H_{hom} from the specific ideals produced by SNOVA, and no released script tests it at a published parameter set.

Theorem A.2 (Conditional finite-cardinality homotopy adaptation). *Assume H_{hom} . Let k be a perfect finite field and put*

$$E_0 = \max \left\{ \max_j \sum_i d_{i,j}, 8B^2 \right\}.$$

If $|k| \geq E_0$, a randomized symbolic-homotopy algorithm whose separating form has a coefficient vector sampled uniformly from k^N returns a zero-dimensional parametrization containing every nonsingular root with probability at least $7/8$. On every run it returns a subset of the nonsingular roots or reports failure. Its arithmetic cost is

$$\tilde{O} \left(BB^+ \left(L + \sum_{i,j} d_{i,j} + N^2 \right) N \right)$$

operations in k . This soft- O theorem does not instantiate its implied constant or polylogarithmic factors. More generally, if $|k| \geq \max_j \sum_i d_{i,j}$ and $|k| > B^2$, the same algorithm succeeds with probability at least $1 - B^2/|k|$.

Proof. Proposition 5 and Algorithm 2 of Safey El Din and Schost give the underlying homotopy construction under their stated hypotheses, with the bad-run behavior described in their Remark 15 [21]. The statement here is a new finite-cardinality adaptation, not a verbatim corollary of that proposition. Its two replacements for the cited characteristic-dependent choices are proved next.

For block j , put $D_j = \sum_i d_{i,j}$ and choose distinct $\alpha_{j,0}, \dots, \alpha_{j,D_j-1} \in k$. The affine moment-curve forms

$$\kappa_{j,u}(X_j) = X_{j,1} + \alpha_{j,u} X_{j,2} + \dots + \alpha_{j,u}^{n_j-1} X_{j,n_j} + \alpha_{j,u}^{n_j}$$

have the same Vandermonde properties as the start forms in the cited proof: any n_j are independent and any $n_j + 1$ are inconsistent. The product start system can be written explicitly. Put

$$s_{i,j} = \sum_{\nu=1}^i d_{\nu,j}, \quad s_{0,j} = 0,$$

and, for $1 \leq i \leq N$, set

$$g_i(X) = \prod_{j=1}^b \prod_{u=s_{i-1,j}}^{s_{i,j}-1} \kappa_{j,u}(X_j).$$

The factor ranges are disjoint. An isolated common zero chooses, for each i , one vanishing factor of g_i ; exactly n_j choices must come from block j . The coefficient in (51) counts these choices, including the $d_{i,j}$ possible factors. The Vandermonde property gives one point for each choice, while inconsistency of any $n_j + 1$ forms prevents additional vanishing factors. The corresponding block-diagonal linear Jacobian is nonsingular, so all B roots are simple. They are enumerated within the same soft-linear cost. The first term in E_0 supplies enough distinct nodes.

The homotopy proof next requires a well-separating linear form. It identifies at most B^2 nonzero vectors on which the coefficient vector must not vanish. The cited proof samples from a one-parameter moment-curve family, which is why its field-size condition includes an additional factor depending on N . Here the coefficient vector is instead sampled uniformly from the full space k^N . For each fixed nonzero vector, its annihilator inside k^N is a proper k -linear subspace (possibly of codimension greater than one), so a uniform coefficient vector annihilates it with probability at most $|k|^{-1}$. Since $|k| \geq 8B^2$, a union bound gives success at least $7/8$.

Under H_{hom} , the remaining Newton lifting, specialization, squarefree cleaning, and zero-dimensional parametrization steps of Algorithm 2 are algebraic over a perfect field and use only field operations and inversion at nonsingular points. Finite fields are perfect, so those correctness and arithmetic arguments carry over. Direct substitution into the original system and a Jacobian-rank test remove every spurious or singular output; hence a run returns only nonsingular roots or reports failure. As in the cited Remark 15, a bad run may return a strict subset, and completeness is not certified at comparable cost. The attack wrapper therefore treats absence of an accepted root as a restart rather than as a proof that none exists. $\square$

The theorem bounds time, not memory. A parametrization can contain $O(B)$ field elements. The soft- O factors, fixed transforms, factorization, filtering, memory traffic, and implementation constants are included later in the route-specific multiplier κ_{hom} .

B Detailed recovery instantiations

This appendix derives the arithmetic ledger behind [Theorem 7.1](#). It first fixes one cost model, then derives the complete systems used as fallbacks and the certified smaller systems used by the optimized route. The final subsection supplies the coefficient-level proofs deferred from [Sections 5](#) and [6](#).

B.1 Arithmetic cost model

All routes below use the same ledger. Pointwise extension-field products use the tower basis and the displayed multiplication recurrences. The unmeasured factor κ_{hom} contains basis conversion, bulk evaluation and interpolation transforms, additions, inversions, factorization, control, memory, and all other hidden costs. Thus every displayed exponent is a multiplication-schedule estimate, not a wall-clock or low-memory bound. The artifact contains a narrower variable-product count, but no theorem uses it, and it performs no production-size solve.

Theorem B.1 (Constructive scalar netlist and tower recurrences). *The circuit model assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, and fan-out. For canonical five-bit representatives, the following circuit-size assignments are constructive:*

	addition	negation	subtraction	multiplication
$\mathbb{F}_{19}$	42	16	58	150
$\mathbb{F}_{19^2}$	84	32	116	692
$\mathbb{F}_{19^4}$	168	64	232	2628

The emitted 150-gate scalar netlist is correct on all $19^2 = 361$ canonical input pairs. The 692 and 2628 extension-field multiplication figures are constructive composition counts from the displayed

primitive recurrences; the release does not emit complete netlists for those two multipliers. The tower representations are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \quad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

They are exactly isomorphic to the manuscript basis $\mathbb{F}_{19}[t]/(t^4 - t - 1)$ via

$$u = 1 + 2t + 15t^2 + 5t^3, \quad v = 4 + 13t + 13t^2 + t^3,$$

and the basis $(1, u, v, uv)$ has determinant $16 \neq 0$ modulo 19.

Proof. The scalar circuit first converts a canonical input $x = (x_0, \dots, x_4)$ into a sign bit s and a four-bit magnitude m . On the valid inputs $0, \dots, 18$, set

$$a = x_1x_3, \quad b = x_2x_3, \quad s = a \oplus b \oplus ax_2 \oplus x_4,$$

$$(m_0, m_1, m_2, m_3) = (x_0 \oplus s, x_1 \oplus s, x_2, x_3 \oplus b).$$

A truth-table check gives $s = 0, m = x$ for $x \leq 9$ and $s = 1, m = 19 - x$ otherwise. The two decoders use 18 gates. The magnitude product and output sign reach gate 80. The product is at most 81, so write it as $\ell + 32h$ with $h \in \{0, 1, 2\}$, replace $32h$ by $13h \bmod 19$, and reduce once. This reaches gate 130. The final 20 gates map the result to r or $-r \bmod 19$ as required by the sign. The released evaluator checks all 361 canonical input pairs against the canonical output.

For $\mathbb{F}_{19^2}$, Karatsuba uses three scalar products and two input sums. The shared output negations give

$$5 \cdot 42 + 2 \cdot 16 + 3 \cdot 150 = 692.$$

For $\mathbb{F}_{19^4}$, three $\mathbb{F}_{19^2}$ products cost $3 \cdot 692$. The two input sums cost $2 \cdot 84$. Multiplication by $1 + u$ costs one scalar addition and one subtraction. The two outputs account for the remaining extension-field operations, giving

$$2 \cdot 84 + 3 \cdot 692 + (42 + 58) + 84 + (84 + 116) = 2628.$$

Direct polynomial arithmetic verifies both tower equations, the displayed coordinates, and the basis determinant. $\square$

Lemma B.2 (Conservative well-separating-form success). *For a square multihomogeneous core with Bézout number B , represented over a finite field E large enough for the finite-cardinality start system, the well-separating-form construction has a set of at most B^2 nonzero forbidden coefficient vectors. A uniform linear form therefore satisfies every separation condition with probability at least*

$$\sigma_E(B) = 1 - \frac{B^2}{|E|}.$$

Conditional on this event and H_{hom} , the finite-cardinality homotopy algorithm returns a zero-dimensional parametrization containing every nonsingular core root, with one-sided error.

Proof. Failure means that the coefficient vector of the linear form annihilates one of at most B^2 nonzero vectors. A uniform coefficient vector annihilates each fixed vector with probability $|E|^{-1}$, so the union bound gives $B^2/|E|$. Pairwise separation alone would count fewer hyperplanes, but it does not cover every condition required by the homotopy parametrization. $\square$

For a core over $A \in \{\mathbb{F}_{19^2}, \mathbb{F}_{19^4}\}$ with Bézout number B and start-node requirement D_{start} , let g_A be 692 or 2628 and define

$$\Gamma_A(B, D_{\text{start}}) = \min_{\substack{e \geq 1: |A|^e \geq D_{\text{start}}, \\ 2e-1 \leq |A|}} \frac{(2e-1)g_A}{1 - B^2/|A|^e}. \quad (53)$$

Evaluation/interpolation at $2e-1$ base-field points gives the number of charged pointwise multiplications in the numerator; the fixed transforms themselves remain in κ_{hom} . [Theorem B.2](#) gives the denominator. The numerical artifact exhausts all admissible e .

B.2 Complete-system fallback

The fallback solves every residual equation. The $\ell = 4$ shapes use an ordinary square over a fresh public slice, while the $\ell = 2$ shapes use all three bidegree families. Both routes apply the accepted-root theorem to the Jacobian of the complete system and filter every returned point through the original verifier.

B.2.1 The ordinary square for $\ell = 4$

Theorem B.3 (Conditional recovery from the complete square system). *Retain the setting of [Theorem 6.2](#), assume $\mathbf{H}_{\text{hom}}$, and suppose $h = K$. Restrict the complete residual map g to the public affine K -dimensional slice and solve all K equations together. On the event $\bar{\eta}_L^U \leq \eta$, a trial contains an accepted root at which the Jacobian of this very system is nonsingular with probability at least*

$$p_{\square} = \frac{a^2}{a + \eta}, \quad a = \alpha - \frac{\eta}{q-1}.$$

If every equation has ordinary degree at most two, then

$$B_{\square} = 2^K, \quad B_{\square}^+ = 2^K \left(1 + \frac{K}{2}\right).$$

Let

$$D_K = \binom{K+1}{2}, \quad L_{\square} = D_K + 2K(D_K + K + 1), \quad H_{\square} = L_{\square} + 2K + K^2.$$

Under $\mathbf{H}_{\text{hom}}$ and over any field satisfying the displayed finite-cardinality conditions, the multiplication work before κ_{hom} is

$$W_{\square} = \frac{\Gamma_A(2^K, 2K)}{p_{\square}} 2^{2K} \left(1 + \frac{K}{2}\right) H_{\square} K. \quad (54)$$

The algorithm is Las Vegas after complete public filtering and assumes no selected Macaulay degree, exact corank, unique root, preferred core, or orbit sweep.

Proof. Because $h = K$, [Theorem 6.2](#) applies to the Jacobian of the complete residual system itself. No equation projection is needed. The ordinary Bézout number of K quadratics in K variables is 2^K . For the companion coefficient, taking no auxiliary term contributes 2^K , while taking it from one of the K factors contributes 2^{K-1} . Therefore $B_{\square}^+ = 2^K + K2^{K-1}$.

The evaluation program forms the D_K quadratic monomials and then evaluates all K affine quadratics. The conditional homotopy bound and [Theorem B.2](#) now give (54). Complete public filtering checks every returned point against the full residual system, affine reconstruction, signature rule, and original verifier. $\square$

Corollary B.4 (Conditional direct-square routes for the six $\ell = 4$ shapes). *For every official $\ell = 4$ shape, a public key that passes the exact quotient and affine-rank checks leaves a fresh ordinary K -dimensional slice in the public-vinegar region. Assume the applicable spectrum inequality and H_{hom} . Applying [Theorem B.3](#), embedding the $\mathbb{F}_{19}$ coefficients into $\mathbb{F}_{19^2}$, and charging the constructive 692-gate multiplication recurrence gives normalized exponents*

$$\boxed{142.60696, \quad 185.45088, \quad 227.56041}$$

on the worse shape at Levels I, III, and V. The corresponding nominal headroom below 143, 207, 272 is

$$\boxed{0.39304, \quad 21.54912, \quad 44.43959}$$

bits before κ_{hom} . The two direct Level-I comparisons survive only for multipliers below approximately 1.313157 and 1.313691, respectively. A factor of two erases both crossings.

Proof. The fresh-slice inequality gives $\dim(V' \cap \ker C_{R,v}) \geq K$ for all six rows, so the required ordinary K -slice exists. The full-domain spectrum theorem applies with $t = 4(v - 2)$ and threshold $\eta_{128}(K, K)$. Substitute $K = 50, 70, 90$ in [\(54\)](#). Exhaustive search over the separator extension degree selects $e = 13, 17, 22$ over $\mathbb{F}_{19^2}$. The ledger then conditions on the exact structural check, divides by the certified spectrum probability, and uses the conservative signature-acceptance bound. $\square$

B.2.2 The zero-offset split for $\ell = 2$

For the three official $\ell = r = 2$ rows put

$$A = \mathbb{F}_{19^2}, \quad m = m_1 = o, \quad M = 4m, \quad K = 3m.$$

Use the zero-offset column relation

$$u_0 = u, \quad u_1 = 0. \tag{55}$$

Proposition B.5 (Exact zero-offset target split). *Let Q_R be the quotient-output matrix for [\(55\)](#). The route applies to a fixed key only after the exact public test rank $\text{rank } Q_R = K$. No probability claim replaces that fixed-key test for the three $\ell = 2$ rows. After a passing test, choose a left inverse G_R and a full-row-rank matrix W_R spanning the left kernel of Q_R . The complete substituted verifier is equivalent to*

$$W_R t = 0, \quad z_{\text{sym}}(u) = G_R t. \tag{56}$$

For a uniform official target, consistency has probability $19^{-(M-K)} = 19^{-m}$. Conditioned on consistency, the residual target is uniform in $\mathbb{F}_{19}^K$, so the attacker can filter message and salt inputs before solving. Charging 2^{32} AXN operations per retained uniform target gives the additive exponents

$$16 \log_2 19 + 32 = 99.96684, \quad 24 \log_2 19 + 32 = 133.95026, \quad 32 \log_2 19 + 32 = 167.93368.$$

Proof. Zero offsets remove the affine and constant terms, leaving $Q_R z_{\text{sym}}(u) = t$. The invertible output change with block rows G_R, W_R gives [\(56\)](#). It preserves uniformity and separates the residual target from the consistency coordinates. Since target filtering precedes the nonlinear solve, its work adds to the solve ledger. $\square$

In one block, (55) has matrix $\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix}$ and is symmetric exactly when $y = 0$. The $n = v + o$ block conditions use disjoint coordinates. Thus the exact density accepted by the Version 2.3 rule is

$$\alpha_n = 19^{-n} \sum_{j=0}^{\lfloor n/4 \rfloor} \binom{n}{j} 18^{n-j}. \quad (57)$$

Choose all solver slices inside the full public-vinegar summand A^v , independently of its fresh internal symmetric coefficients. By Theorem 6.3, the fresh exponent is $t = 2(v - 1)$.

The complete two-block system. Take an A -linear slice of dimension $s = 3m/2$. Its base-field dimension is $h = 2s = K$, and the complete quotient system has m equations in each bidegree family

$$(2, 0), \quad (1, 1), \quad (0, 2).$$

It is therefore square and includes every residual equation.

Theorem B.6 (Conditional complete two-block $\ell = 2$ recovery). *On the event $\bar{\eta}_L^U \leq \eta_{128}(K, K)$, a consistent target-coset trial contains an accepted full-Jacobian root with probability at least*

$$p_{\text{comp}}^{(2)} = \frac{a_2^2}{a_2 + \eta_{128}(K, K)}, \quad a_2 = \alpha_n - \frac{\eta_{128}(K, K)}{18}. \quad (58)$$

The exact multihomogeneous quantities are

$$B_m^{(2)} = 2^{2m} \binom{m}{m/2}, \quad (59)$$

$$(B_m^{(2)})^+ = B_m^{(2)} \left(1 + 2m + \frac{m^2}{m+2} \right). \quad (60)$$

Assuming $\mathbf{H}_{\text{hom}}$, finite-cardinality homotopy over $\mathbb{F}_{192}$ recovers the roots without choosing a Macaulay degree or a preferred subsystem. Frobenius descent and complete public filtering reject every invalid candidate. Including the additive target filter and the 2^{-128} spectrum normalization gives the following exponents before $\log_2 \kappa_{\text{hom}}$:

$$132.97067, \quad 183.79463, \quad 233.84399$$

for Levels I, III, and V.

Proof. Apply Theorem 6.2 with $h = K$ and the accepted density from (57). Over the splitting field, each base form contributes one equation to each diagonal family and one to the cross family. Hence the Bézout number is

$$[t_0^s t_1^s] (2t_0)^m (t_0 + t_1)^m (2t_1)^m.$$

The cross factor must contribute $m/2$ powers to each block, proving (59).

For the companion coefficient, the term with no auxiliary variable contributes $B_m^{(2)}$. Deleting a diagonal factor contributes $2^{2m-1}(\binom{m}{m/2} + \binom{m}{m/2+1})$, while deleting a cross factor contributes $B_m^{(2)}$. Summing and using $\binom{m}{m/2+1}/\binom{m}{m/2} = m/(m+2)$ proves (60). Finally, (53) accounts for the separating form, and every later operation is an exact filter. $\square$

For the complete two-block system, define

$$\begin{aligned}
D_s &= \binom{s+1}{2}, & C_s &= (s+1)^2, \\
L_{\text{comp}} &= 2D_s + s^2 + 4m(D_s + s + 1) + 2mC_s, \\
H_{\text{comp}} &= L_{\text{comp}} + 6m + K^2.
\end{aligned} \tag{61}$$

On a good spectrum event, the multiplication work before κ_{hom} is

$$W_{\text{comp}}^{(2)} = \frac{\Gamma_{\mathbb{F}_{19^2}}(B_m^{(2)}, 2s)}{p_{\text{comp}}^{(2)}} B_m^{(2)} (B_m^{(2)})^+ H_{\text{comp}} K. \tag{62}$$

Divide by the exact spectrum-event density and add the target-generation ledger.

B.2.3 All-nine complete-system costs

Corollary B.7 (All-nine complete-system fallback). *Consider any published $q = 19$ Version 2.3 parameter shape. Suppose the public key and column relation pass the exact rank checks, the complete-system bound $\bar{\eta}_L^U \leq \eta_{128}(K, K)$ holds, and H_{hom} holds for the invoked system. Then the complete-system algorithm checks every candidate against the affine reconstruction, signature rule, and original verifier, so it never returns an invalid signature. The algorithm does not choose a Macaulay degree or assume a unique root, predicted corank, or predetermined Jacobian minor.*

The worst direct costs at Levels I, III, and V are

$$\boxed{142.60696, \quad 185.45088, \quad 233.84399}. \tag{63}$$

The nine direct rows and their optimized counterparts appear together in Table 5. These values include the idealized target-generation ledger from Section 5 and precede the unmeasured multiplier κ_{hom} .

Proof. For $\ell = 4$, Theorem B.4 gives the ordinary squares. For $\ell = 2$, Theorem B.6 gives the two-block squares. Both use Theorem 6.2, the separating-form bound, and complete public filtering. Substitution of the nine published shapes gives Table 5; taking the largest direct entry at each level gives (63). $\square$

B.3 Certified smaller systems and optimized costs

The complete systems above give the fallback. A smaller route uses fewer linear combinations of the residual equations. The algorithm first checks the leading Jacobian determinant with public data. It solves the smaller system only when this exact test passes and otherwise returns to the complete system.

B.3.1 Certified projection and complete-square fallback

The public test checks that the highest homogeneous part of the selected system's Jacobian determinant is not the zero polynomial. In the applications below, evaluating that polynomial at one fixed public point suffices.

Theorem B.8 (Certified smaller system with complete-square fallback). *Fix a public construction for which a common structural and spectrum event $\mathcal{S}$ makes the complete-system fallback valid. Suppose:*

1. when the exact public determinant test passes, the smaller system has expected ledger cost at most W_f ;
2. when the test fails, the complete fallback has expected work at most W_F ;
3. under the idealized XOF-transcript distribution, the test fails with probability at most δ , while $\Pr[\mathcal{S}^c] \leq \varepsilon$.

For every fixed key in $\mathcal{S}$, the algorithm tests the smaller system and otherwise uses the complete fallback. Under the idealized transcript distribution, its conditional expected work obeys

$$\mathbb{E}[W \mid \mathcal{S}] \leq W_f + \min\left\{1, \frac{\delta}{1 - \varepsilon}\right\} (W_F - W_f)_+. \quad (64)$$

If an independent exact structural test succeeds with probability p_{str} , the transcript-model-normalized ledger divides the right side by $p_{\text{str}}(1 - \varepsilon)$. Without independence one may instead use any proved lower bound on $\Pr[\mathcal{S}]$.

Proof. For a fixed key, the determinant test is a certificate rather than a probabilistic premise. A passing key uses the smaller system, while a failing key uses the complete system. Both branches apply the full public filters.

For the transcript-model cost, let F be the event that the determinant test fails. Then

$$\Pr[F \mid \mathcal{S}] \leq \frac{\Pr[F]}{\Pr[\mathcal{S}]} \leq \min\{1, \delta/(1 - \varepsilon)\}.$$

If $q = \Pr[F \mid \mathcal{S}]$ and $W_F \geq W_f$, the branch average is $W_f + q(W_F - W_f)$. If $W_F < W_f$, always using the fallback only lowers the cost. This gives the positive-part expression. The last statement divides by the density of idealized transcripts that reach the solver. $\square$

B.3.2 The smaller Level-I $\ell = 2$ system

For the $\ell = 2$ rows, retain $A = \mathbb{F}_{19^2}$, $m = o$, $K = 3m$, and the accepted density α_n from (57). For $s \leq m$, select s diagonal equations and solve one eigenblock of A^s . A descended full root reconstructs the other block by Frobenius. The core is a square system of s ordinary quadratics, with

$$B_s = 2^s, \quad B_s^+ = 2^s(1 + s/2).$$

At a fixed nonzero public point, the fresh leading row of each selected public form is surjective onto $(A^s)^\vee$. Sequential exposure therefore gives

$$p_{\text{Jac}}^{(2)}(s) \geq \prod_{j=1}^s (1 - \rho_{\text{byte}}^{2j}) > 0.9970003329. \quad (65)$$

A nonzero leading determinant has degree at most s over A , so its singular point density is at most $s/19^2$. Put $\eta_s = \eta_{128}(2s, 3m)$ and $a_s = \alpha_n - s/19^2$. The singleton and second-moment arguments then give the following lower bound for an accepted root at which this core is nonsingular:

$$p_s = 19^{2s-3m} \max\left\{a_s - \eta_s, \frac{a_s^2}{a_s + \eta_s}\right\}. \quad (66)$$

At Level I, set $s = m = 16$ and run the exact leading-determinant test. A passing key uses the one-eigenblock system. A failing key uses the complete two-block system from Theorem B.6. The fallback theorem gives normalized exponent

$$\boxed{132.18807}. \quad (67)$$

The diagonal-only and complete-only exponents in the reported ledger are 132.18928 and 132.97067. The adaptive value is slightly smaller than the first rounded value because the ledger applies the common spectrum normalization once and combines the two branches before taking logarithms. At Levels III and V, the complete system is already faster.

The separate Just Guess sensitivity for the Level-I $\ell = 2$ row appears in [Section C](#). It is not used by the main recovery theorem.

B.3.3 The smaller $\ell = 4$ channel systems

Let $A = \mathbb{F}_{19^4}$ and let $L \cong A^s$ be the public slice. Under the Frobenius-distance normal form, each public base form supplies an A -valued degree-two channel and an A -valued degree-20 channel. Write their highest homogeneous parts as

$$F_{i,0}^{\text{top}}(x) = C_{i,0}(x, x), \quad F_{i,1}^{\text{top}}(x) = C_{i,1}(x, x^{19}).$$

Lemma B.9 (Fresh channel decomposition and row surjectivity). *Restriction of a fresh symmetric $\mathbb{F}_{19}$ -bilinear form to the injected public-vinegar copy of L , followed by the Frobenius-distance transform, is surjective onto*

$$\text{Sym}_A^2(L^\vee) \oplus \text{Bil}_A(L, L^{(19)}; A).$$

For every fixed $0 \neq x_\star \in L$, the map from one public form's fresh coefficients to the pair of leading Jacobian rows

$$(DF_{i,0}^{\text{top}}(x_\star), DF_{i,1}^{\text{top}}(x_\star)) \in L^\vee \oplus L^\vee$$

is surjective over A .

Proof. After extending scalars, the four diagonal blocks form one Frobenius orbit. They descend to an arbitrary symmetric A -bilinear form. The four adjacent blocks form a disjoint orbit and descend to an arbitrary bilinear form between L and its Frobenius twist. Since the two coefficient sets are disjoint, the map onto the two channels is surjective.

Formal differentiation gives

$$DF_{i,0}^{\text{top}}(x_\star)[h] = 2C_{i,0}(x_\star, h), \quad DF_{i,1}^{\text{top}}(x_\star)[h] = C_{i,1}(h, x_\star^{19}).$$

Evaluation at nonzero $x_\star$ is surjective for arbitrary bilinear forms. For the symmetric channel, given $\lambda \in L^\vee$, choose $\phi(x_\star) = 1$ and use the symmetric form

$$\frac{1}{2}(\phi \otimes \lambda + \lambda \otimes \phi - \lambda(x_\star)\phi \otimes \phi).$$

Its evaluation at $x_\star$ is λ , which proves row surjectivity. $\square$

Choose a whole coordinates of the degree-two channel and b whole coordinates of the degree-20 channel, with $a + b = s$. The determinant test is exact on the public key. The following transcript-model bound does not assume that an invertible linear change preserves independence under the biased reduction modulo 19.

Lemma B.10 (Transcript-model failure bound for the Jacobian test). *At the fixed public test point, let $J \in A^{s \times s}$ be the selected leading-Jacobian matrix, where $A = \mathbb{F}_{19^4}$. Then*

$$\Pr[\det J = 0] \leq \frac{|A|^s - 1}{|A| - 1} \rho_{\text{byte}}^{4s}. \quad (68)$$

For the cost-minimizing values $s = 10, 14, 18$, the failure bounds are at most $3.55350 \cdot 10^{-5}$, $6.56029 \cdot 10^{-5}$, and $1.21113 \cdot 10^{-4}$, respectively.

Proof. If J is singular, its right kernel contains a projective direction $[y] \in \mathbb{P}^{s-1}(A)$. Fix $[y]$. Row surjectivity implies that forcing all s selected rows to annihilate y imposes s independent equations over A . These become $4s$ independent affine equations over $\mathbb{F}_{19}$ in fresh public symbols. The maximum-atom bound gives probability at most ρ_{byte}^{4s} for this direction.

There are $(|A|^s - 1)/(|A| - 1)$ projective directions. A union bound proves (68). Although weaker than the full-rank product for uniform field elements, this bound applies to the biased product distribution in the idealized transcript model. $\square$

Conditional on passing the determinant test, put

$$a_{a,b} = \alpha - \frac{a + 19b}{19^4}.$$

The sharpened core-root probability is

$$p_{a,b,s} = 19^{4s-K} \frac{a_{a,b}^2}{a_{a,b} + \eta_{128}(4s, K)}. \quad (69)$$

The exact Bézout data for this core are

$$B = 2^a 20^b, \quad B^+ = B(1 + a/2 + b/20).$$

Define its straight-line evaluation cost by

$$\begin{aligned} L_{a,b,s} = & \mathbf{1}_{a>0} \left[\binom{s+1}{2} + 2a \left(\binom{s+1}{2} + s + 1 \right) \right] \\ & + \mathbf{1}_{b>0} [6s + s^2 + 2b(s^2 + 2s + 1)], \end{aligned}$$

Under $\mathbf{H}_{\text{hom}}$, the multiplication ledger for a key that passes the determinant test is

$$\frac{\Gamma_{\mathbb{F}_{19^4}}(B, 2a + 20b)}{p_{a,b,s}} BB^+ (L_{a,b,s} + 2a + 20b + s^2)s. \quad (70)$$

Exhaustive enumeration of all 35, 63, and 99 admissible profiles gives

$$(s; a, b) = (10; 5, 5), \quad (14; 7, 7), \quad (18; 9, 9)$$

at Levels I, III, V. Conditional on the exact structural test and the stated spectrum/model events, and assuming $\mathbf{H}_{\text{hom}}$, the worse-shape smaller-system exponents before $\log_2 \kappa_{\text{hom}}$ are 126.260, 167.038, and 207.273.

The artifact regenerates every numerical input for these profiles. The paper makes no numerical claim for a sixteen-orbit or output-size-optimized branch.

B.3.4 Combining the certified branches

For $\ell = 2$, the Level-I test selects one eigenblock when its determinant certificate passes and otherwise uses the complete two-block square. The complete square is already cheaper at Levels III and V. For $\ell = 4$, the projective Jacobian bound of Theorem B.10 controls the fast branch, and a failed test returns to the complete system of Theorem B.4. The artifact combines the two branch costs before taking logarithms, charges the union of their spectrum events, and normalizes only by the probability of the common spectrum event.

The exact ledger gives the following sufficient conditions for the unmeasured homotopy multiplier:

$$\log_2 \kappa_{\text{hom}}^{(I)} < 10.811927, \quad \log_2 \kappa_{\text{hom}}^{(III)} < 23.205374, \quad \log_2 \kappa_{\text{hom}}^{(V)} < 38.156013.$$

These inequalities are rounded downward. The analysis does not assert a particular value for any of these multipliers.

B.4 Proofs for square-slice preparation and spectrum bounds

This subsection collects the coefficient-level proofs used in [Sections 5](#) and [6](#). The corresponding statements and proof ideas remain in the main body.

Proof of [Theorem 5.1](#). The family in (24) has $m_1 d^2$ members. It is stable under right multiplication by A because, for every $\zeta \in A$,

$$(w^\top \Sigma_\xi P_i \Sigma_\eta) \Sigma_\zeta = w^\top \Sigma_\xi P_i \Sigma_{\eta\zeta},$$

which is another member of the span. Every offset cross term contains one copy of w . If w occurs on the right, symmetry and commutativity move it to the left. Thus every row of $F_{R,v}$ lies in U_w , and right stability puts every orbit row there as well.

The complete right orbit in (25) makes its row space an A -vector space, so $d \mid \rho_F$. Its first block is $F_{R,v}$, hence its kernel kills every offset-linear feature. If x is in the kernel and $\zeta \in A$, each $S^a \zeta$ is an $\mathbb{F}_q$ -linear combination of the chosen basis powers. The same orbit rows therefore kill $\Sigma_\zeta x$, proving that the kernel is A -linear. Rank-nullity over A gives (26). $\square$

Proof of [Theorem 5.2](#). Every structured offset lies in W , so each offset-linear coefficient contains one copy of w . Symmetry moves that copy to the left when needed. The coefficient therefore reads only native matrix entries with at least one index in W . Right multiplication by A preserves this coordinate block, as do the deterministic orbit and slice constructions.

The native $V' \times V'$ entries come from disjoint XOF symbols and remain unread. They are therefore independent of the selected slice in the stated model. This argument must stay in the native coordinates because the biased byte-reduction distribution is not invariant under an arbitrary A -linear basis change. $\square$

Proof of [Theorem 5.3](#). Choose t pivot columns. Condition on all nonpivot variables. Expose the pivot variables in echelon order; at each step at most one value can satisfy the next independent equation, and that value has probability at most ρ_{byte} . Multiplication gives the claim. $\square$

Proof of [Theorem 5.4](#). First consider failure of the affine source rank. If $\text{rank } C_{R,v} < M - K$, some nonzero $\lambda \in \mathcal{Q}_R^\vee$ annihilates $C_{R,v}$. In particular, $(\lambda \circ C_{R,v})|_{V'} = 0$. For fixed $[\lambda]$, this is an affine system of rank at least $d(v-1)$ in the independent native variables $\mathbf{X}_{W,V'}$; [Theorem 5.3](#) bounds its probability by $\rho_{\text{byte}}^{d(v-1)}$. Scalar multiples define the same event, and $\mathbb{P}(\mathcal{Q}_R^\vee)(\mathbb{F}_q)$ has $(q^{M-K} - 1)/(q - 1)$ points. A union bound proves (28).

Next consider failure of the projection to V' . By [Theorem 5.1](#), $H_{R,v}^F$ is A -linear. A nonzero intersection with $W \oplus O_{\text{pub}}$ must therefore contain an A -line $\mathfrak{l}$. For $\mathfrak{l} = W$, the coefficient-rank hypothesis and [Theorem 5.3](#) give probability at most $\rho_{\text{byte}}^{m_1 \binom{d+1}{2}}$. For every other A -line the analogous bound is $\rho_{\text{byte}}^{m_1 d^2}$. The $(1+o)$ -dimensional A -space $W \oplus O_{\text{pub}}$ contains $(|A|^{1+o} - 1)/(|A| - 1)$ lines, exactly one of which is W ; a union bound gives (29). Finally, the kernel of the projection $H_{R,v}^F \rightarrow V'$ is precisely $H_{R,v}^F \cap (W \oplus O_{\text{pub}})$.

A final union bound combines the source-rank and projection failures without assuming independence. Direct substitution gives the stated numerical bound. $\square$

Proof of [Theorem 6.3](#). Fix $y \neq 0$ and $b \neq 0$. Choose a fresh public base form whose corresponding symmetric-square coefficient c is nonzero. Writing $c = \sum_\nu a_\nu \odot b_\nu$, the fresh symmetric-matrix coefficient in the scalar derivative $b \cdot T_y(x)$ is

$$\Xi_{c,\bar{y}}(x) = \sum_\nu ((a_\nu \bar{y}) \odot (b_\nu x) + (b_\nu \bar{y}) \odot (a_\nu x)).$$

We first bound the kernel of this coefficient map. If $x \notin A\bar{y}$, project onto the cross summand between the disjoint lines $A\bar{y}$ and Ax . After identifying both lines with A , this projection is the ordinary symmetrization of c . It is nonzero because symmetrization is injective on $\text{Sym}^2(A)$ in odd characteristic. Hence the kernel is contained in $A\bar{y}$, and the coefficient map has rank at least $\dim_{\mathbb{F}_q} D - d = t$.

The event $b \in \ker T_y^\top$ forces a rank- t affine system in independent fresh symbols. Conditioning on the nonpivot symbols and applying the maximum-atom bound to the pivots proves (38).

Finally, the dual-kernel identity gives

$$q^{-\text{rank } T_y} = q^{-K} |\ker T_y^\top| = q^{-K} \left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^\top\}} \right).$$

Its expectation is at most $q^{-K} + (1 - q^{-K})\rho_{\text{byte}}^t$. Summing over $y \neq 0$ proves (39). The excess above the baseline $(q^h - 1)q^{-K}$ is nonnegative, so Markov's inequality proves (40). The two stated choices of D have dimensions $d(v - 1)$ and dv . $\square$

An unused fixed-skew format check. A deterministic alternative is also available: for the official matrix

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \quad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets with $(v_1)_{b,0} - (v_0)_{b,1} = 1$ in every block. Then each block has fixed nonzero skew 1, so every root passes the rejection rule. This fixed-skew route is a useful separate algebraic check but is not needed for the reported recovery theorem.

C A Boolean-ledger Level-I sensitivity via Just Guess

The homotopy ledger deliberately leaves κ_{hom} explicit. This section asks whether the row that determines the adaptive Level-I maximum, $(v, o, \ell, r) = (48, 16, 2, 2)$, admits a more concrete estimate. We combine the exact zero-offset target split of Theorem B.5 with the underdetermined-MQ transformation of May, Ostuzzi, and Ressler [18]. Their analysis reduces a random underdetermined MQ system to guessed variables, a short tree of univariate quadratics, a linear solve, and check equations. It explicitly uses a random-system regularity model for the linearization step. We retain that premise rather than turning it into an unconditional statement.

Write the quotient map in its two Frobenius channels as

$$D : A^{64} \longrightarrow A^{16}, \quad H : A^{64} \longrightarrow \mathbb{F}_{19}^{16}, \quad A = \mathbb{F}_{19^2}.$$

Here D is the diagonal channel and H is the Hermitian cross channel. The retained consistent target is (t_D, t_H) . We additionally reject the event $t_D = 0$, of probability 19^{-32} under the exact uniform target sampler. Conditioned on consistency and $t_D \neq 0$, the pair (t_D, t_H) remains exactly uniform over the retained target set.

The two channel families are distinct linear images of the same native byte-reduced public coefficients. They are therefore *not independent*. The argument below uses an exact conditional atom bound for the official $\ell = 2$ channel transform instead.

C.1 The streamed diagonal solver

Apply Just Guess to the $m = 16$ equations $D(x) = t_D$ in $n = 64$ variables over A , with

$$k = 5, \quad p = 6, \quad m - k - p = 5.$$

The published structural inequalities hold:

$$2k + p = m, \quad 2(n - m) = 96 \geq 6(2m - 2k - p - 1) = 90,$$

$$n - 1 = 63 \geq (m - k - p - 1)(m - k + 2) = 52.$$

The public transformation gives coordinates $(u, z) \in A^{16} \times A^{48}$. Fixing z changes only linear and constant terms in the transformed square system. We stream an A^8 family of z assignments. For each assignment, the solver enumerates all $|A|^5$ guesses, follows every returned quadratic branch, solves the remaining 5×5 linear system, and checks the five withheld equations. Surviving points are tested against $H(x) = t_H$, the format rule, affine reconstruction, and the unmodified verifier.

Let $\mathcal{R}$ denote the event that this computation supplies at least $19^{16}/4$ distinct sign-canonical diagonal roots. This is a concrete, transcript-checkable candidate-abundance premise motivated by, but not proved by, the published Just Guess random-system analysis. No reduced-size or production transcript in the release establishes $\mathcal{R}$. We write κ_{JG} for the expected multiplier needed to obtain $\mathcal{R}$. It includes transformation and linearization retries, candidate-abundance failures, the negligible format and $t_D = 0$ losses, and any candidate-processing cost beyond the explicit reserve below. Unlike κ_{hom} , it does not hide a generic soft- O algebraic solver, but it remains an unmeasured premise. If $\mathcal{R}$ is unreachable for a fixed system and transformation family, then $\kappa_{\text{JG}} = \infty$ and the comparison gives no crossing.

C.2 Conditional cross-channel filtering

For $x \in A^{64}$ put $\Psi(x) = xx^{19\top}$. If $\Psi(x) = \Psi(y)$ for nonzero x, y , then $y = \lambda x$ with $\lambda\lambda^{19} = 1$. If also $D(x) = D(y) = t_D \neq 0$, then $\lambda^2 = 1$, so $y = \pm x$. Retaining one representative of each sign pair therefore makes the cross tensors distinct.

Lemma C.1 (Exact conditional channel atom). *Use the native-coordinate idealized XOF-transcript model, in which one field symbol has raw integer weight 14 for residues $0, \dots, 8$ and 13 for residues $9, \dots, 18$. Condition on the complete diagonal-channel coefficient family D . For every native coefficient block and every nonzero $\mathbb{F}_{19}$ -linear functional of its remaining Hermitian coefficient, every conditional atom has probability at most*

$$\rho_{\text{cond}} = \frac{49}{829} < 0.059108. \tag{71}$$

Conditioned on D , different native blocks and different base forms remain independent.

Proof. Diagonalize the official matrix $S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}$ over $A = \mathbb{F}_{19}[u]/(u^2 + 1)$. Its eigenvalues are $8 \pm 2u$, with eigenvectors $(1, 13 \pm u)$. For a native symmetric diagonal block $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the diagonal coefficient and Hermitian coefficient are

$$D = (a + 7b + 16c) + (2b + 7c)u, \quad H = a + 7b + 18c.$$

For an off-diagonal native block $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, they are

$$D = (a + 13b + 13c + 16d) + (b + c + 7d)u,$$

$$H = (a + 13b + 13c + 18d) + (18b + c)u.$$

These maps act separately on disjoint native blocks. Exact enumeration of all 19^3 diagonal tuples with their product weights gives maximum conditional atom $49/829$. Exact enumeration of all 19^4 off-diagonal tuples and all 20 projective nonzero linear functionals on the A -valued Hermitian coefficient gives the smaller maximum $169246/2971565$. The artifact reproduces both rational values directly from the displayed formulae. Conditioning separately on the D -image of each independent native block preserves blockwise and base-form independence. $\square$

Lemma C.2 (Conditional cross-channel second moment). *Let C be any set of $T = c19^{16}$ sign-canonical solutions of $D(x) = t_D$, where $t_D \neq 0$ and $0 < c \leq 1$. Condition on the entire diagonal channel and on any solver transcript determined by it. For a uniform independent target t_H ,*

$$\Pr[\exists x \in C : H(x) = t_H \mid D] \geq \frac{c}{1 + c(19\rho_{\text{cond}})^{16}}. \quad (72)$$

For $c = 1/4$, this lower bound is $0.0961344407\dots$

Proof. Let $N = |\{x \in C : H(x) = t_H\}|$. Uniformity of t_H gives $\mathbb{E}N = c$. For distinct $x, y \in C$, the Hermitian tensors differ. For each of the sixteen independent base forms, the equality $H_i(x) = H_i(y)$ is a nonzero linear equation in the Hermitian block coefficients. Condition on all blocks except one on which this functional is nonzero. By [Theorem C.1](#), its conditional probability is at most ρ_{cond} . Independence across the sixteen base forms gives

$$\Pr[H(x) = H(y) \mid D] \leq \rho_{\text{cond}}^{16}.$$

Consequently

$$\mathbb{E}N(N-1) \leq T(T-1)19^{-16}\rho_{\text{cond}}^{16} < c^2(19\rho_{\text{cond}})^{16}.$$

Cauchy–Schwarz gives $\Pr[N > 0] \geq (\mathbb{E}N)^2/\mathbb{E}[N^2]$, proving the claim. $\square$

C.3 Finite Boolean-gate instantiation

We use the constructive 692-gate multiplication recurrence and 84-gate adder count over A . A monic quadratic over A is solved by a fixed Tonelli–Shanks schedule. The artifact checks all 361 possible discriminants and obtains

$$C_{\text{root}} = 16,886 \quad \text{AXN gates.} \quad (73)$$

Instantiating the published expected Just Guess operation formulas at $(n, m, k, p) = (64, 16, 5, 6)$ gives, per guess,

$$2684.089 \text{ extension multiplications,} \quad 2621.087 \text{ extension additions,} \quad 6 \text{ quadratic-root calls.}$$

Rounding upward and adding one million gates for comparisons, address logic, and branch control gives

$$C_{\text{guess}} = 3,179,584 \quad \text{AXN gates.} \quad (74)$$

We declare a reserve of 2^{40} gates per z assignment for candidate filtering and verifier calls. This is a ledger allowance, not a theorem that it dominates every possible candidate population; any excess is included in κ_{JG} . Target sampling, the public transformation, and all rank preflights are polynomial-sized additive terms.

Proposition C.3 (Conditional finite-gate Level-I sensitivity). *Under the published Just Guess expected-operation model, the Boolean ledger for one A^8 -streaming trial is*

$$W_{\text{JG}}^{\text{trial}} = |A|^{8+5}C_{\text{guess}} + |A|^8 2^{40} < 2^{132.046523}. \quad (75)$$

On event $\mathcal{R}$, Theorem C.2 with $c = 1/4$ gives an expected cross-channel restart multiplier below 10.403. Consequently the success-adjusted Boolean-gate ledger is

$$\log_2 W_{\text{JG}} < 135.425325 + \log_2 \kappa_{\text{JG}}. \quad (76)$$

It is below the Level-I reference 143 whenever

$$\log_2 \kappa_{\text{JG}} < 7.574675. \quad (77)$$

For orientation only, a factor-16 regularity reserve gives an exponent below 139.425325 and more than 3.574675 bits of remaining margin. A separate capped-tree schedule follows at most 63 quadratic nodes and 64 leaves per guess and aborts any branch whose residual linear system is not unique. Its trial exponent is below 137.713504; after the same cross-channel restart it is below $141.092307 + \log_2 \kappa_{\text{JG}}^{\text{cap}}$. Thus this strictly bounded-work sensitivity remains below 143 when $\log_2 \kappa_{\text{JG}}^{\text{cap}} < 1.907693$. It is reported as a sensitivity check, not as a proof of the capped candidate event.

Proof. There are $|A|^8$ streamed assignments and $|A|^5$ guesses per assignment. Substituting the scalar circuit, tower recurrence counts, and quadratic-root schedule into the published expected-operation formulas gives (75); the artifact recomputes the integer ledger before taking logarithms. On $\mathcal{R}$, take any $19^{16}/4$ sign-canonical candidates. The reciprocal of (72) is $10.4020993\dots$, adding 3.378803 bits. The remaining regularity, transformation, candidate-collection, and excess filtering multipliers are, by definition, included in κ_{JG} . Every candidate is checked by the original verifier, so all failure modes are one-sided. $\square$

Interpretation. This proposition replaces the homotopy soft- O multiplier on the row setting the adaptive Level-I maximum by (i) an explicit Boolean ledger for the published expected-operation schedule and (ii) the specific, unestablished but transcript-checkable event that the Just Guess run yields enough diagonal roots. It does not provide a complete circuit for every control path or prove candidate abundance. A complete, instrumented fixed-key implementation would estimate κ_{JG} . The two $\ell = 4$ Level-I rows still retain κ_{hom} in their homotopy ledgers.

D Streamed-XL and low-state analyses

The complete-square homotopy route supplies a conditional time bound under $\mathbf{H}_{\text{hom}}$, but it does not supply a low-memory implementation. This section records a complementary branch whose matrices can be streamed with linear vector state. Its correctness is exact once a finite quotient certificate is produced; its numerical exponent retains a production selected-degree premise. The two branches therefore answer different questions rather than competing for one headline.

D.1 Macaulay matrices and the low-state branch

XL turns quadratic equations into a larger linear system by multiplying them by selected monomials. A multigraded version tracks the degree in each variable block and can produce a smaller matrix for block-structured systems [8, 9]. Recovery does not require a one-dimensional kernel. Commuting multiplication matrices can also describe all roots through FGLM and finite-field factorization [22, 23]. The analysis below states its additional production-degree assumption.

D.2 Eigenblocks and multidegrees

Let Ω split the minimal polynomial of S , and let $e_0, \dots, e_{\ell-1}$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is A -linear of A -dimension s , then

$$L \otimes \Omega = L_0 \oplus \dots \oplus L_{\ell-1}, \quad \dim_{\Omega} L_a = s.$$

The unordered power-pair features are an invertible recombination of the unordered eigenblock features

$$x^{\top} \Sigma_{e_a} P_i \Sigma_{e_b} x, \quad 0 \leq a \leq b < \ell.$$

They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$. Translation creates lower-degree terms only in the same one or two blocks. One homogenizer per block therefore preserves these exact multidegrees. This is the grading used by the streamed-XL branch.

Proof of the eigenblock statement. The power basis and the idempotent basis of $A \otimes \Omega \cong \Omega^{\ell}$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between unordered power-pair and unordered eigenblock coordinates. The idempotent e_a projects $L \otimes \Omega$ onto L_a . A diagonal feature uses one block twice; an off-diagonal feature is bilinear in two blocks. Translation and homogenization preserve the claimed supports. $\square$

D.3 Four-block special XL

For an A -linear $\ell = 4$ slice of A -dimension s , scalar extension gives four blocks of s variables. Each unordered pair of blocks carries m_1 independent quadratic equations. After adding one homogenizer per block, the Hilbert series used by the published special-XL model is

$$H_{m_1, s}(\mathbf{t}) = \frac{\prod_{a=1}^4 (1 - t_a^2)^{m_1} \prod_{1 \leq a < b \leq 4} (1 - t_a t_b)^{m_1}}{\prod_{a=1}^4 (1 - t_a)^{s+1}}. \quad (78)$$

For a multidegree $\mathbf{d} = (d_1, \dots, d_4)$, the monomial-column count is

$$M_s(\mathbf{d}) = \prod_{a=1}^4 \binom{s + d_a}{d_a}. \quad (79)$$

A negative coefficient of (78) is the selected-degree trigger in the cited model [8]. It predicts that a planted instance has the finite quotient needed for extraction; the theorem below makes the operator and state accounting exact but does not prove that production instances reach such a quotient at that degree.

The exact row count is

$$\begin{aligned} R_s(\mathbf{d}) = & m_1 \sum_{a=1}^4 \binom{s + d_a - 2}{d_a - 2} \prod_{c \neq a} \binom{s + d_c}{d_c} \\ & + m_1 \sum_{1 \leq a < b \leq 4} \binom{s + d_a - 1}{d_a - 1} \binom{s + d_b - 1}{d_b - 1} \prod_{c \notin \{a, b\}} \binom{s + d_c}{d_c}, \end{aligned} \quad (80)$$

where a binomial coefficient with a negative lower argument is zero.

Proposition D.1 (Streaming the special-XL operator). *Let $A_{\mathbf{d}}$ be the Macaulay matrix at multidegree $\mathbf{d}$, with $M = M_s(\mathbf{d})$ columns and $R = R_s(\mathbf{d})$ rows. Every row and transpose-row product can be generated from the public quadratic coefficients using $O(s^2)$ scratch space.*

Over an extension $E/\mathbb{F}_{19^4}$, choose a random diagonal $\Delta \in E^{R \times R}$ and put

$$B = A_{\mathbf{d}}^T \Delta A_{\mathbf{d}}.$$

Then

$$\Pr[\ker B \neq \ker A_{\mathbf{d}}] \leq M/|E|.$$

A product by B is evaluated by one pass over the streamed rows and uses $O(M)$ resident field-vector state. Scalar Wiedemann over $E = \mathbb{F}_{19^8}$ requires fewer than

$$16R(s+1)^2 M \tag{81}$$

signed $\mathbb{F}_{19^4}$ multiply-accumulates per trial. A conservative cubic preconditioner over $\mathbb{F}_{19^{12}}$ gives the Level-I bound $32R(s+1)^2 M$.

Proof. The two sums in (80) count multipliers of the diagonal and off-diagonal equation families. A homogenized quadratic row has at most $(s+1)^2$ terms, so rows and transpose products are generated on demand.

Factor $A_{\mathbf{d}} = UV$ at rank r . If $U^T \Delta U$ is nonsingular, then $\ker(A_{\mathbf{d}}^T \Delta A_{\mathbf{d}}) = \ker V = \ker A_{\mathbf{d}}$. By Cauchy–Binet, $\det(U^T \Delta U)$ is a nonzero multilinear polynomial of total degree r in the diagonal entries. Schwartz–Zippel gives failure at most $r/|E| \leq M/|E|$.

Streaming first computes the row inner product and then accumulates the scaled row. In the quadratic extension, the sparse dot product and accumulation use at most four base-extension products per row term and the diagonal multiplication uses three more. Fewer than $3M$ normal-operator products generate the scalar Wiedemann sequence and reconstruct one null vector. Rounding the resulting constant gives (81); the cubic-extension calculation is analogous. $\square$

D.4 Certified extraction beyond exact corank one

A one-dimensional Macaulay kernel is convenient but is not required for soundness.

Theorem D.2 (Recovery from a finite commuting border quotient). *Fix a target and slice over a finite field Ω , and let $I \subseteq \Omega[x_1, \dots, x_h]$ be the ideal of the residual equations. Let $\mathcal{B}$ be a finite divisor-closed monomial set containing 1. Suppose the Macaulay row space supplies, for every border monomial $\beta \in \partial\mathcal{B}$, a certified relation*

$$g_{\beta} = \beta - \sum_{b \in \mathcal{B}} c_{\beta,b} b \in I.$$

Construct the formal multiplication matrices $M_1, \dots, M_h$ on $\Omega\mathcal{B}$. If they commute, then the border relations generate a zero-dimensional ideal $J \subseteq I$, $\mathcal{B}$ is a basis of $\Omega[x]/J$, and every root of I occurs among the roots enumerated from the multiplication tables. FGLM conversion costs $O(h|\mathcal{B}|^3)$ field operations once the tables are available. Complete residual and public verification removes every extra point.

Proof. Commutativity makes $x_i \mapsto M_i$ an algebra homomorphism. Divisor closure identifies each basis monomial with its coordinate vector, and every border relation acts as zero. The border-division theorem therefore gives unique normal forms in $\text{span}_{\Omega} \mathcal{B}$, so the relations form a border basis of a zero-dimensional ideal J . Because every relation is a Macaulay row combination from I , we have $J \subseteq I$ and hence $V(I) \subseteq V(J)$. Standard FGLM conversion and finite-field factorization enumerate the geometric points of J [22, 23]. $\square$

A quotient of dimension larger than one is therefore a correctness fallback, not a free extension of the scalar-Wiedemann exponent. Obtaining all border relations, block-nullspace work, and resident multiplication-table state must fit the displayed ledger or be charged separately.

D.5 Three-plus-one completion

The fourth eigenblock can be completed linearly after solving the first three. Let E_{ab} be the m_1 equations supported on blocks a, b .

Theorem D.3 (Exact three-plus-one completion). *Suppose a certified first-stage solver returns a finite set containing every root of the six families E_{ab} with $1 \leq a \leq b \leq 3$. For each partial root $p = (x_1, x_2, x_3)$, substitute into E_{14}, E_{24}, E_{34} and obtain*

$$A_p x_4 = b_p, \quad A_p \in \Omega^{3m_1 \times s}.$$

Discard inconsistent systems. If $\text{rank } A_p = s$, solve for the unique fourth block and retain it exactly when $E_{44} = 0$. If a consistent system has rank below s , declare the trial incomplete unless a separately priced procedure enumerates all completions. Whenever no incomplete case occurs, the retained set contains every root of the complete four-block system. Final descent and verification make the procedure Las Vegas.

Proof. Any complete root restricts to a first-stage root. At that partial root its true fourth block satisfies the displayed linear system. On a trial that declares no incomplete case, every consistent system encountered has full column rank. Hence the true fourth block is the unique completion, and the remaining diagonal family checks it. Conversely every retained point satisfies all ten families. $\square$

D.6 Status of the low-state branch

The streamed operator, row count, and correctness fallbacks above remain useful complementary machinery. No production numerical profile is reported because the artifact does not regenerate a selected multidegree, success-probability bound, and state figure from one checker. A numerical frontier would require both that profile generator and a production border-basis or solve transcript.

E A finite fixed-key spectrum-certificate format

The idealized XOF-transcript theorem is not required for a statement about a concrete key. This appendix gives a finite exact certificate format for the aggregate spectrum. No such production-key spectrum certificate is shipped in the release.

Put $N_U = \dim_{\mathbb{F}_q} U$. Choose a basis $y_0, \dots, y_{h-1}$ of a fixed slice L and write

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \dots, Y_{h-1}]^{K \times N_U}.$$

For $0 \leq j < h$, use the disjoint normalized projective chart $Y_0 = \dots = Y_{j-1} = 0$, $Y_j = 1$, and put $R_j = \mathbb{F}_q[Y_{j+1}, \dots, Y_{h-1}]$. Let $T^{(j)}$ be the resulting matrix. For $t \geq 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \quad (82)$$

Put $n_{j,\leq t} = \dim_{\mathbb{F}_q}(R_j/I_{j,t})$, $n_{j,\leq -1} = 0$, and

$$N_t = \sum_{j=0}^{h-1} (n_{j,\leq t} - n_{j,\leq t-1}). \quad (83)$$

Proposition E.1 (Projective rank-histogram certificate). *The integer N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ for which $\text{rank } T_y = t$. Consequently*

$$\bar{\eta}_L^U = (q-1) \sum_t N_t q^{-t}. \quad (84)$$

Reduced Gröbner bases with their standard monomial sets, or polynomial reduction transcripts proving the unit-ideal and quotient-dimension claims, form machine-checkable fixed-key certificates.

Proof. The normalized charts contain exactly one representative of every projective $\mathbb{F}_q$ -direction. The field equations identify

$$R_j / \langle Y_i^q - Y_i : i > j \rangle \cong \prod_{a \in \mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all generators vanish. The minors in (82) vanish exactly when $\text{rank } T^{(j)}(a) \leq t$. Thus $n_{j, \leq t}$ counts rank-at-most- t points in chart j , and differencing gives (83). Every projective direction has $q-1$ nonzero scalar representatives and rank is invariant under base-field scaling, proving (84). $\square$

A stronger certificate that $\text{rank } T_y \geq t_0 + 1$ for every nonzero y consists of unit-ideal transcripts for all I_{j, t_0} . The complete histogram is more flexible: a small exceptional low-rank locus can be charged exactly instead of controlling the whole pencil by its minimum rank.

F Dimension formulas and exact numerical rows

For a Version 2.3 row,

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell+1}{2}.$$

For a full affine lift with $\text{rank } C_{R,v} = M - K$,

$$\dim \ker C_{R,v} = N_0 - M + K.$$

For a structured lift whose complete offset orbit has rank $m_1 \ell^2$,

$$\dim_A H_{R,v}^F = n - m_1 \ell.$$

The direct ordinary fresh-slice lower bound for $\ell = 4$ is

$$\dim(V' \cap \ker C_{R,v}) \geq 4(v-1) - (M-K).$$

Table 5: Exact unit-leading exponents underlying the rounded main text. Every row conditions on its exact public quotient/rank preflights, the applicable spectrum/model events, and H_{hom} ; the $\ell = 4$ rows additionally condition on the fixed-layout structural preflight. The values exclude $\log_2 \kappa_{\text{hom}}$. “Direct” solves the complete square; “adaptive” is the conditional mixture of an exact smaller-system determinant test and the complete-square fallback.

Level	Parameters	Direct	Adaptive	Adaptive headroom
I	(28, 5, 4, 4)	142.606960	128.246278	14.753722
I	(48, 16, 2, 2)	132.970670	132.188072	10.811928
I	(28, 4, 4, 5)	142.606374	128.245747	14.754253
III	(40, 7, 4, 4)	185.450879	171.616678	35.383322
III	(72, 24, 2, 2)	183.794625	183.794625	23.205375
III	(38, 5, 4, 5)	185.450045	171.615857	35.384143
V	(50, 9, 4, 4)	227.560406	214.558321	57.441679
V	(96, 32, 2, 2)	233.843987	233.843987	38.156013
V	(52, 6, 4, 6)	227.559359	214.557277	57.442723

All values charge the 150, 692, and 2628-gate multiplication circuits as appropriate and use the conservative B^2 forbidden-vector separator. For $\ell = 4$, the Jacobian-preflight density uses the projective right-kernel union bound, not an independence claim after a biased linear transform.

G Arithmetic-circuit certificate details

The scalar multiplier consumes two little-endian five-bit canonical integers in $\{0, \dots, 18\}$ and returns the canonical five-bit representation of their product modulo 19. Its 150 gates are two-input AND, XOR, and XNOR gates with free constants, wires, fan-out, and structural sharing. The released netlist is tested on all 361 valid input pairs. The SHA-256 digest of the canonical compact JSON payload containing exactly `ninputs`, `gates`, and `outputs` is

e1708a32c1475ad14295c2b34fed976fb09577b127810d29bf237db18aabc131

for the emitted balanced-netlist representation. This is an internal payload identifier, not the bitwise digest of the surrounding ledger file.

The tower recurrences are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \quad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

Three-product Karatsuba at each level, together with the exact linear operations stated in [Theorem B.1](#), gives constructive recurrence counts 692 and 2628. The artifact verifies the tower identities, the basis determinant, and the recurrence arithmetic; it does not emit or exhaustively test complete extension-field netlists.

These circuits normalize only the pointwise multiplication schedule. They are not lower bounds and do not price the whole solver. Additions, inversions, factorization, bulk evaluation/interpolation transforms, filtering, control, indexing, communication, and memory remain in κ_{hom} .

H Reproduction and package contents

The source package is self-contained. Its root contains the complete paper source, bibliography, compiled PDF, build instructions, and hashes. The `artifact/` directory contains:

- a standalone generator and separately coded consistency checker for the all-nine conditional homotopy ledger;
- the complete 150-gate scalar multiplier netlist and tower checks;
- the Just Guess expected-operation and capped ledgers, the quadratic-root check, and the exact conditional Frobenius-channel atom enumeration;
- a public-key correspondence harness for one official Level-I KAT; and
- a non-planted zero-offset end-to-end composition test at the unofficial shape $(2, 1, 2, 2)$, with fresh-key forgery and rank stress checks.

A clean reproduction is:

```
sha256sum -c SHA256SUMS
make regenerate
make verify
make
```

The all-nine generator assigns no random probability to the fixed outer map: its $\ell = 4$ ledgers condition on the exact public structural preflight. The checkers are deterministic, require Python 3.10 or newer plus NumPy for the two correspondence tests, and do not require network access. The paper build uses only the files in the package. The supplied **SHA256SUMS** records every distributed source and pinned input; generated TeX build files are excluded.